

Open and Operational Day-Ahead Electricity Price, Load, Solar, and Wind Forecasting

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Abstract

In the European day-ahead electricity market, bids for each quarter-hour of the next day must be submitted before the auction closes at 12:00. Market participants use price forecasts to prepare these bids, so a forecast is only operational if all of its inputs are available at that time. Many studies use the load, solar, and wind forecasts published by the transmission system operators as price-model inputs. The operators publish these forecasts to meet a legal reporting obligation and do not disclose how they are produced. The solar and wind forecasts are, in addition, only published after 12:00, so they could not be used in real operation.

This thesis develops an open and operational pipeline that produces its own load, solar, and wind forecasts from open data and uses them to forecast day-ahead prices for the German-Luxembourg market. It asks how accurate these forecasts are compared with those of the system operators, whether they improve the price forecast, and how the accuracy of the price forecast changes with information sets that grow hourly from 07:00 to 12:00. Over a February to July 2026 test period, the proposed load forecast reduces the error of the system operators' forecast by 36.0%, while the proposed solar and wind forecasts are less accurate but available in time. Using these forecasts lowers the price forecast error by 11.2%. Forecasts made later in the morning become gradually more accurate, but the only statistically significant gain comes at 12:00, when the price of the Austrian day-ahead auction, which closes earlier, is already available. The results show that open load, solar, and wind forecasts improve operational price forecasts even where they are less accurate than those of the system operators. Beyond these results, the thesis provides a transparent alternative to the forecasts of the system operators, whose methods are not disclosed. The pipeline uses only open data and is publicly released, and its forecasts are submitted every day to the open Energy-Arena benchmark, where their accuracy can be verified independently beyond the test period. Instead of relying on black boxes, anyone can inspect, rerun, and reuse these benchmarked forecasts for their own purposes.

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List of Abbreviations

API	Application Programming Interface
aFRR	automatic Frequency Restoration Reserve
DE-LU	Germany-Luxembourg bidding zone
DWD	Deutscher Wetterdienst
ENTSO-E	European Network of Transmission System Operators for Electricity
EXAA	Energy Exchange Austria
FCR	Frequency Containment Reserve
GW	Giacomini-White
ICON	Icosahedral Nonhydrostatic weather model
ICON-D2	Icosahedral Nonhydrostatic weather model, high-resolution German domain
MAE	Mean Absolute Error
MaStR	Marktstammdatenregister
mFRR	manual Frequency Restoration Reserve
MTU	Market Time Unit
OBTF	Operational Boosted-Tree Forecasting
RMSE	Root Mean Squared Error
RQ	Research Question
SDAC	Single Day-Ahead Coupling
SHAP	SHapley Additive exPlanations
TSO	Transmission System Operator
UTC	Coordinated Universal Time

List of Symbols

Symbol	Meaning
$d, d - 1$	Delivery day and previous calendar day.
$q, \mathcal{Q}_d$	Market time unit and set of market time units on delivery day d .
t	Forecast creation time or cutoff.
$x_{d,q}, \hat{x}_{d,q d-1}$	Realized value of a variable x in market time unit q of delivery day d , and its forecast created on day $d - 1$.
$\mathcal{F}_{d-1}^t$	Information available at cutoff t on day $d - 1$.
$P_{d,q}, \hat{P}_{d,q d-1}$	Realized DE-LU day-ahead price and corresponding point forecast.
$L_{d,q}, S_{d,q}, W_{d,q}$	Realized load, solar, and wind generation.
$G_{d,q}^{\text{ren}}$	Realized renewable generation (solar plus wind).
$L_{d,q d-1}^{\text{ENTSOE}}, S_{d,q d-1}^{\text{ENTSOE}}, W_{d,q d-1}^{\text{ENTSOE}}$	ENTSO-E day-ahead load, solar, and wind forecasts.
$\hat{L}_{d,q d-1}, \hat{S}_{d,q d-1}, \hat{W}_{d,q d-1}$	OBTF load, solar, and wind forecasts.
$\hat{G}_{d,q d-1}^{\text{ren}}$	OBTF renewable-generation forecast (solar plus wind).
$\widehat{RL}_{d,q d-1}$	OBTF residual-load forecast.
$\hat{P}_{d,q d-1}^{\text{base}}$	Base price forecast with the ENTSO-E load forecast.
$\hat{P}_{d,q d-1}^{\text{gen}}$	Price forecast with $\hat{L}_{d,q d-1}$, $\hat{G}_{d,q d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q d-1}$ instead of the ENTSO-E load forecast.
$\hat{P}_{d,q d-1}^{\text{gen}, -\text{wea}}$	Price forecast $\hat{P}_{d,q d-1}^{\text{gen}}$ without raw weather features.
N_{d-1}, N_{d-7}	Previous-day and previous-week naive price forecasts.

1 Introduction

Day-ahead electricity price forecasts support short-term electricity market decisions [1, 2]. In the coordinated European day-ahead market, orders must be submitted before gate closure at 12:00 CET/CEST¹ on the day before delivery [3, 4]. A price forecast for this setting must therefore be created before this deadline. The same timing rule also matters in historical backtests, because using information that was not yet observable at the forecast creation time would introduce data leakage.

Load, solar, and wind generation are central inputs for such price forecasts. Demand affects scarcity, while solar and wind generation reduce the residual demand that has to be served by conventional plants [5, 6]. Through the merit-order effect, generation with low short-run marginal costs can displace more expensive units and reduce spot prices [7]. For this reason, empirical price forecasting models often include load, renewable generation, weather, and calendar information [1, 2, 8]. Some studies bring weather information into the price model directly. For example, Ludwig et al. [9] use a broad set of German weather variables, and Sgarlato and Ziel [10] use meteorological forecasts such as wind, irradiation, cloud cover, and temperature to forecast German electricity prices.

However, several price-forecasting studies for the Germany-Luxembourg bidding zone (DE-LU) use day-ahead load forecasts published through the European Network of Transmission System Operators for Electricity (ENTSO-E) Transparency Platform as price-model inputs [11, 12, 13, 14, 15], and several also use the ENTSO-E-published solar and wind forecasts instead of direct weather variables [11, 12, 14]. These published forecasts compress weather-sensitive system conditions into expected load, solar, and wind generation, so a price model that relies on them implicitly assumes that they are accurate, observable in time, and sufficiently complete. They are, however, effectively black boxes: mandatory transparency data reported by system operators through undisclosed models and information sets rather than reproducible specifications [16, 17]. How good these forecasts are compared with forecasts that are transparent and reproducible from open data, and what they are worth for price forecasting, can therefore only be judged by benchmarking them.

Timing is the second problem, and it decides whether a published forecast can be used at all. For load, the ENTSO-E day-ahead forecast is public before gate closure and can enter forecast runs conducted after its publication [16]. The ENTSO-E solar and wind forecasts, by contrast, are used as price-model inputs in several DE-LU studies [11, 12, 14] but are not publicly observable before the 12:00 gate closure, with a publication requirement no later than 18:00 on day $d - 1$ [18, 17, 16]. For an operationally valid backtest, each input must be observable at the forecast creation time. These forecasts can therefore serve as quality benchmarks but not as pre-gate-closure inputs. This is the operational gap addressed here, which this thesis fills with solar and wind forecasts generated from open data before gate closure.

Because admissibility depends on public observability at the forecast creation time, the information set is itself a central design choice. This thesis defines a strict operational information set for each hourly forecast creation time from 07:00 to 12:00 on day $d - 1$. These sets become richer

¹All clock times in this thesis refer to Central European Time or Central European Summer Time, the local time of the DE-LU market.

towards gate closure as newer weather runs, more recent realized system data, reserve-market auction results, and the public load forecast become observable.

These considerations lead to three research questions (RQ1 to RQ3):

- RQ1:** How do fully transparent, open-data-based forecasts of load, solar, and wind generation compare with the corresponding ENTSO-E forecasts?
- RQ2:** How does an explicit representation of load and renewable-generation forecasts affect electricity price forecasts?
- RQ3:** How does electricity price forecast performance change with richer information sets over time?

The following hypotheses make these questions testable:

- H1:** The self-generated load forecast has lower or comparable forecast errors than the ENTSO-E load forecast, while the self-generated solar and wind forecasts have higher errors than the corresponding ENTSO-E forecasts.
- H2:** Adding explicit self-generated load and renewable-generation forecasts improves electricity price forecast quality, relative to a baseline price model that uses the published ENTSO-E load forecast and represents renewable generation only through raw meteorological inputs.
- H3:** Price forecast accuracy improves as the forecast creation time moves closer to gate closure and the admissible information set expands during the morning before delivery.

The intuition behind H1 is an asymmetry in information. For load, the public forecast is available before gate closure [16], so the self-generated forecast can correct it with newer weather runs and more recent realized load. For solar and wind, no public forecast is available before gate closure that could be corrected in this way. The self-generated forecasts must rely on open weather and capacity data alone, whereas the ENTSO-E forecasts are produced by the system operators with undisclosed models and inputs and may be issued as late as 18:00 on day $d - 1$ [16, 17], so higher errors are expected for these two targets.

The central contribution of this thesis is a set of fully operational forecasting models for load, solar, and wind generation based exclusively on open data, referred to as operational boosted-tree forecasting (OBTF). The models are released as part of an open pipeline that is ready for operational use and that others can rerun and reuse. Their forecasts, together with the resulting price forecasts, are submitted every day to the open Energy-Arena benchmark [18], so their accuracy can be verified independently and continuously, also beyond the test period. In contrast to the ENTSO-E forecasts, whose methods are not disclosed, the inputs and construction of these models are fully open. By construction, each model is admissible before gate closure, so it can be used where the corresponding ENTSO-E forecast cannot, as for solar and wind. The benchmarking in RQ1 further shows the OBTF load model to be even more accurate than the ENTSO-E load forecast. The price forecast is one application that demonstrates their operational value.

The remainder of the thesis is structured as follows. Chapter 2 reviews related work, Chapter 3 describes the data, models, and evaluation design, Chapter 4 presents the results, and Chapters 5 and 6 discuss and conclude the thesis.

2 Literature Review

This review does not attempt a comprehensive survey of electricity price forecasting, for which recent reviews are available [2, 1]. It focuses instead on the question that motivates this thesis: how do existing day-ahead price models represent their inputs, and is that representation admissible in an operational, pre-gate-closure setting? Table 1 is the central output of the review. Adapting the feature overview of Eiser and Kleinebrahm [15], it records for each study which input features enter the price model, whether direct weather is used, whether all inputs are observable before gate closure, and whether the study is open and reproducible, and contrasts these with this thesis.

Day-Ahead Price Forecasting. Weron [1] provides a broad review of electricity price forecasting and emphasizes the volatility, seasonality, and short-run dependence of spot prices. Lago et al. [2] extend this view by reviewing modern algorithms, proposing an open benchmark for day-ahead electricity price forecasting, and showing that strong empirical designs usually combine price lags, calendar variables, and exogenous system information. Kath and Ziel [19] study quarter-hourly electricity price forecasts and show that more accurate forecasts at this resolution can create economic value. The quarter-hourly resolution of the DE-LU price target follows from the market design rather than from a modeling choice. Together, these contributions motivate the use of lagged prices, calendar variables, and simple naive price benchmarks in this thesis.

Load and Renewable Generation in Price Models. Load, solar, and wind generation are fundamental drivers of the price, because they describe the physical supply-and-demand state of the power system [6, 7]. Sensfuß et al. [7] document the merit-order effect for Germany and show how renewable feed-in can reduce spot prices by shifting more expensive conventional generation out of the market. Pape et al. [5] use a fundamental supply-and-demand model of the German day-ahead and intraday market to examine how far load, renewable generation, and the conventional supply stack explain observed price variations, while Trebbien et al. [20] use explainable machine learning to show that German electricity prices depend on interacting drivers beyond a simple merit-order view. Most directly, Maciejowska et al. [8] study the role of load, wind, and solar generation information in day-ahead electricity price forecasting and show that including these forecast variables, and improving their accuracy, yields more accurate price forecasts. Their comparison is between price models with and without these forecasts, rather than against a model that already uses direct weather variables. Whether self-generated load, solar, and wind forecasts still improve a price model that already includes raw weather therefore remains open and is tested in RQ2.

A separate line of research brings weather variables directly into the price model. Ludwig et al. [9] forecast German spot prices using historical prices and a large set of weather variables, while Sgarlato and Ziel [10] show that meteorological forecasts can improve German electricity price forecasts beyond the day-ahead horizon. These papers show that direct weather-to-price representations are feasible. They also make the representation question explicit: weather can enter a price model directly, or it can enter indirectly through load, solar, and wind forecasts. Using forecasts as this indirect representation raises two questions that this thesis addresses. First, the forecasts must be accurate, and RQ1 benchmarks the OBTF load, solar, and wind forecasts against the ENTSO-E forecasts. Second, they must capture enough of the weather-driven state to

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stand in for raw weather, which RQ2 examines.

Forecasting Load, Solar, and Wind Generation. Generating these forecasts is itself a forecasting task with an established literature for each target. For load, Hong and Fan [21] show why calendar effects, recent load observations, and weather variables are central predictors. For photovoltaic generation, Antonanzas et al. [22] identify radiation, cloud cover, temperature, and plant characteristics as key inputs, while Foley et al. [23] review wind power forecasting methods and emphasize the role of wind speed, wind direction, atmospheric conditions, and turbine locations.

Public Forecasts and Operational Timing. Many DE-LU price forecasting models rely on system forecasts published on the ENTSO-E Transparency Platform. In their review, Eiser and Kleinebrahm [15] find that these models commonly use the day-ahead load forecast and, in several cases, the day-ahead solar and wind forecasts, while direct weather variables are not listed as a separate input category. Table 1 compares the day-ahead price-forecasting studies most directly related to this thesis. It places these models, which use the published ENTSO-E load and renewable-generation forecasts as price-model inputs, next to the weather-direct studies and the operational pipeline of Eiser and Kleinebrahm [15], and records their input features, use of direct weather, pre-gate-closure admissibility, and openness. For price modeling, these published load and renewable-generation forecasts act as summary variables that compress weather, calendar, and system information into expected load, solar, and wind generation, so a model that uses them and omits direct weather treats them as the relevant compression of weather information.

In hindsight, every published forecast is available in the data, including forecasts that were not yet observable when the price forecast would have been made. A valid backtest must therefore apply the same restriction as an operational forecast, namely that every input is observable before the 12:00 gate closure [3]. Under the transparency regulation the public load forecast is published no later than two hours before gate closure and is observed around 10:30, so it is admissible before gate closure [16, 17]. The forecast values are published, but the models that produce them are not disclosed. Their errors can be measured against the realized values, but an error level alone does not show how good a forecast is compared with alternatives. RQ1 therefore benchmarks these forecasts against forecasts built from open data. The day-ahead price model of Uniejewski [13], for instance, relies on price lags, calendar variables, and this admissible load forecast, without any renewable-generation forecasts. The ENTSO-E solar and wind day-ahead forecasts, by contrast, are not publicly observable before gate closure, with a publication requirement no later than 18:00 on day $d - 1$ [18, 16], so this thesis uses them only as benchmarks. Faced with this constraint, Eiser and Kleinebrahm [15] substitute spatially resolved weather for the unavailable renewable-generation forecasts rather than forecasting renewable generation explicitly, and the Energy-Arena benchmark [18] enforces the same pre-gate-closure timing across contributors. The reviewed studies do not systematically examine the trade-off between forecasting earlier and using later-arriving information before gate closure. This thesis addresses both gaps. In RQ1 it generates its own OBTF load, solar, and wind forecasts from information available before gate closure and benchmarks them against the corresponding ENTSO-E forecasts. In RQ3 it then examines how price forecast accuracy changes as the forecast creation time moves toward gate closure.

Weather and Spatial Information. Weather forecasts are the main input for load, solar, and wind forecasts [21, 22, 23]. The Icosahedral Nonhydrostatic (ICON) model framework is

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described by Zängl et al. [24], and the Deutscher Wetterdienst (DWD) provides operational weather forecast products as open data [25]. Open-Meteo is used as an additional open weather application programming interface (API) in the pipeline [26]. The weather fields have to be weighted by where demand and generation are located. Population data from Eurostat therefore weight the weather for the OBTF load model, and installed renewable capacity from the Marktstammdatenregister (MaStR) weights it for the OBTF solar and wind models [27, 28].

Table 1: Input features, pre-gate-closure admissibility, and openness of selected German and DE-LU day-ahead price-forecasting studies compared with this thesis. A check mark denotes a reported property, a dash an absent or unreported one. The *ENTSO-E* and *Self-generated* columns mark the use of published ENTSO-E forecasts or of forecasts produced from admissible data. *Pre-GC*: all inputs observable before the 12:00 gate closure. *Data*, *Code*: released dataset or pipeline and public code. *Benchmark*: results reported on Energy-Arena. RES: renewable energy sources. EUA: EU emission allowances. ^aStates a 12:00 cutoff but uses the post-closure ENTSO-E RES forecast. ^bBias-corrects the published ENTSO-E forecasts. ^cCorrection of the published ENTSO-E load forecast.

Study	General inputs				Load / RES forecast				Operational & openness			
	Price lags	Direct weather	Fuel/EUA	Calendar	ENTSO-E load	ENTSO-E RES	Self-generated load	Self-generated RES	Pre-GC	Data	Code	Benchmark (Energy-Arena.org)
Trebbien et al. [11]	✓	–	✓	–	✓	✓	–	–	– ^a	–	–	–
Marcjasz et al. [12]	✓	–	✓	✓	✓	✓	–	–	–	✓	✓	–
Hilger et al. [14]	✓	–	–	–	✓	✓	–	–	–	–	–	–
Maciejowska et al. [8]	✓	–	–	✓	✓	✓ ^b	–	–	–	–	–	–
Uniejewski [13]	✓	–	–	✓	✓	–	–	–	✓	–	–	–
Ludwig et al. [9]	✓	✓	–	–	–	–	–	–	✓	–	–	–
Sgarlato and Ziel [10]	✓	✓	–	–	–	–	–	–	✓	–	–	–
Eiser and Kleinebrahm [15]	✓	✓	–	✓	✓	–	–	–	✓	✓	✓	✓
This thesis	✓	✓	–	✓	–	–	✓ ^c	✓	✓	✓	✓	✓

Operational Benchmarking. The operational requirements emphasized in this review are brought together in the Energy-Arena operational forecasting benchmark [18]. Energy-Arena hosts day-ahead forecasting challenges for the DE-LU and Austrian markets, with separate load, solar, wind, and price challenges as well as challenges tied to different forecast creation times. It scores each submission against those of the other contributors on the realized market outcome under a common gate-closure deadline, so that all participants face the same operational information constraint and their forecasts can be compared directly on a public leaderboard. It thereby turns the pre-gate-closure setting studied here into an external, leakage-controlled testbed. This thesis submits its OBTF load, solar, and wind forecasts and its resulting price forecasts to the benchmark at different cutoff times, where they remain publicly accessible and reproducible and can be reused by others, for example as members of forecast ensembles. Energy-Arena is thus where the elements reviewed above come together in practice: transparent open-data inputs, forecasts produced under an admissible information set, a fixed pre-gate-closure timing, and open, reusable outputs.

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Summary and Positioning. That load, solar and wind generation, weather, calendar variables, and price lags matter for day-ahead price forecasting is well established. Three less obvious points motivate this thesis. First, the representation of load and renewable generation matters, not only their inclusion. Maciejowska et al. [8] show that how load and renewable generation are predicted, and how accurate those predictions are, changes the price forecast. Second, how weather should be represented for a price model is not systematically studied. Weather usually enters only indirectly, through published load and renewable-generation forecasts, so the model inherits whatever those forecasts capture, and the studies that do use weather directly neither compare it against that indirect representation nor test whether the load and renewable-generation forecasts add value on top of direct weather. This thesis takes up that representation question, comparing a weather-based price model with one that adds the OBTF load, solar, and wind forecasts and testing, through the weather-free variant, whether those forecasts can substitute for raw weather (RQ2).

Third, and most importantly, the operational information set is often left underspecified. A cutoff time alone does not define the experimental setup. What matters is which inputs are actually observable at that time, yet the reviewed studies often do not establish which inputs are admissible by then. Some therefore include inputs that are published only after gate closure, such as the ENTSO-E renewable-generation forecasts [11, 12, 14], while informative and clearly admissible signals, such as early exchange prices [29], are left out without comment. The first opens the setup to information leakage, the second discards admissible information, and both follow from not establishing which inputs are admissible. This thesis instead defines the admissible information set explicitly and tracks how it grows through the morning up to gate closure, so that every comparison uses only information that was genuinely available at the forecast creation time.

None of the reviewed operational DE-LU studies generates dedicated load, solar, and wind forecasts from admissible information and uses them as price-model inputs, and none systematically examines how price forecast quality changes with the forecast creation time before gate closure. This thesis addresses both gaps, and does so transparently: it uses open data, releases its full pipeline as open source, and submits its forecasts to the Energy-Arena. These properties are contrasted across the reviewed studies in Table 1.

This gap leads directly to the methodology. The thesis generates OBTF load, solar, and wind forecasts from admissible information, compares them with the corresponding ENTSO-E day-ahead forecasts, and tests whether they add value to the day-ahead price model. Finally, it evaluates how price forecast performance changes as the forecast creation time moves from 07:00 toward 12:00.

3 Methodology

Figure 1 summarizes the complete setup of the thesis.

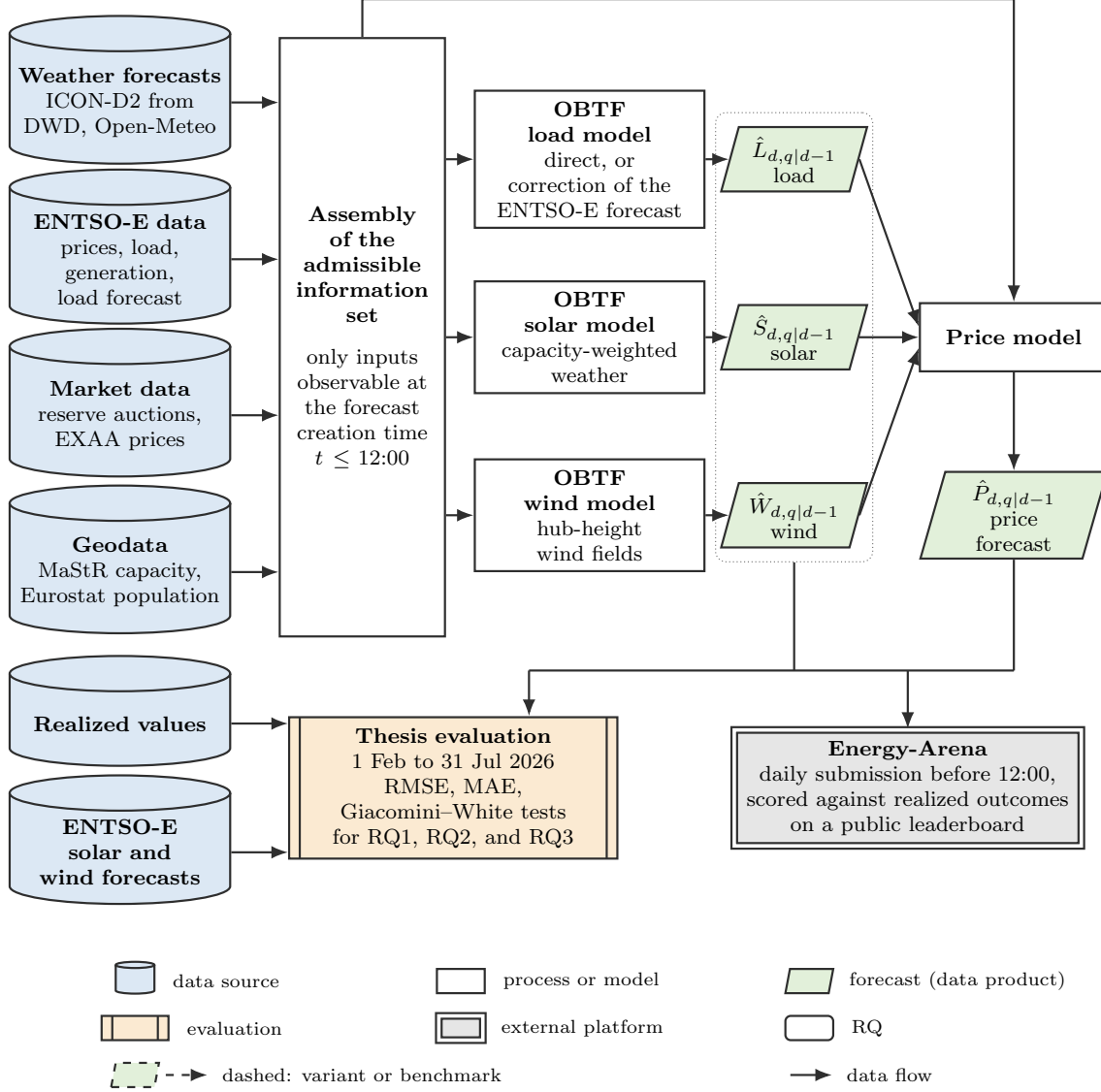


Figure 1: Overview of the complete setup of the thesis, from the open data sources through the forecasting models to the evaluation and the daily submission. The legend applies to all process diagrams in this thesis.

In this setup, open data sources are assembled into the admissible information set at the forecast creation time t . This set feeds the OBTF load, solar, and wind models, and the price model receives their forecasts together with further inputs from the same set. All models are gradient-boosted trees. In RQ3, the pipeline is rerun for each forecast creation time. The OBTF and price forecasts are evaluated on the test period for RQ1 to RQ3. Data published only after gate closure, namely the realized values and the ENTSO-E solar and wind forecasts, are used only for evaluation and

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never as model inputs.

The pipeline is deployed operationally on the Energy-Arena platform [18], which hosts separate forecasting challenges for load, solar, wind, and the DE-LU day-ahead price, as well as challenges tied to different forecast creation times. The pipeline submits forecasts to each challenge every day before the corresponding submission cutoff, which is the 12:00 gate closure for the day-ahead price and correspondingly earlier for the challenges tied to earlier forecast creation times. This deployment provides a live check that every model respects the operational information cutoff and that the setup can be run end to end under realistic timing constraints. The final configuration of each model in the pipeline is summarized in Appendix A2.

3.1 Problem Formulation

This thesis considers the operational point forecasting problem of predicting quarter-hourly day-ahead electricity prices in the DE-LU bidding zone. Let $P_{d,q}$ denote the realized day-ahead price for delivery day d and market time unit (MTU) q . On regular days, the day consists of 96 quarter-hourly MTUs, and the forecast for delivery day d is created on day $d - 1$, the day before delivery. It must be available before the day-ahead gate closure at 12:00 [3]. For a forecast creation time t on day $d - 1$, the price forecasting task is written as

$$\hat{P}_{d,q|d-1} = f(\mathcal{F}_{d-1}^t, q), \quad q \in \mathcal{Q}_d,$$

where $\hat{P}_{d,q|d-1}$ is the point forecast for delivery day d , $\mathcal{Q}_d$ is the set of MTUs, and $\mathcal{F}_{d-1}^t$ is the information set available at time t . Unless stated otherwise, the information cutoff is set to the latest feasible pre-gate-closure time, $t = 12:00$, and the forecast is computed and submitted slightly before the 12:00 deadline.

The central idea is to generate forecasts of load, solar, and wind generation inside the same operational information set and then use them as explicit inputs to the price model. The full notation is collected in the List of Symbols. The OBTF forecasts are

$$\begin{aligned} \hat{L}_{d,q|d-1} &= g^{\text{load}}(\mathcal{F}_{d-1}^t, q), \\ \hat{S}_{d,q|d-1} &= g^{\text{solar}}(\mathcal{F}_{d-1}^t, q), \\ \hat{W}_{d,q|d-1} &= g^{\text{wind}}(\mathcal{F}_{d-1}^t, q). \end{aligned}$$

The OBTF solar and wind forecasts are combined into the OBTF renewable-generation forecast

$$\hat{G}_{d,q|d-1}^{\text{ren}} = \hat{S}_{d,q|d-1} + \hat{W}_{d,q|d-1},$$

because the price responds to the total renewable feed-in that lowers residual load rather than to the split between the two sources (Section 3.4.5). Both are still forecast separately by their own models. Together with the OBTF load forecast, this defines the OBTF residual-load forecast,

$$\widehat{RL}_{d,q|d-1} = \hat{L}_{d,q|d-1} - \hat{G}_{d,q|d-1}^{\text{ren}}.$$

The base price forecast, without the OBTF load, renewable-generation, and residual-load fore-

casts, is denoted $\hat{P}_{d,q|d-1}^{\text{base}}$. The price forecast with OBTF load and renewable-generation inputs, namely $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$, is denoted $\hat{P}_{d,q|d-1}^{\text{gen}}$. These objects define the comparisons examined in the experimental design (Section 3.5).

3.2 Data

3.2.1 Target Variable and Temporal Scope

The target variable is the realized DE-LU day-ahead auction price $P_{d,q}$, measured in EUR/MWh. The German day-ahead market changed from hourly to quarter-hourly market clearing on 1 October 2025 [30]. Post-transition observations therefore contain genuine 15-minute price variation, whereas pre-transition prices only reflect hourly market clearing and are treated as historical proxy information if they are used to extend a rolling training window. In that case, each hourly price is expressed at quarter-hourly resolution by applying it to the four quarter-hours of that hour, since pre-transition clearing set a single price for the whole hour.

For model development and evaluation, the post-transition sample is split into a development period and a held-out out-of-sample test period. The split, the rolling estimation scheme, and their justification are defined in Section 3.5.

All timestamped data are represented in the local market timezone Europe/Berlin. To keep the model target dimension constant across delivery days, daylight-saving transition days are mapped to the fixed 96-column MTU representation used by the models: repeated autumn intervals are averaged into their MTU, while missing spring intervals are filled by within-day linear interpolation. The models and the reported evaluation use this fixed 96-unit representation, so predicted and realized prices are always compared on the same grid. An operational submission to Energy-Arena, which scores each delivery day on its actual MTU grid of 92, 96, or 100 units on 23-, 24-, and 25-hour days, maps the 96-unit forecast back to those units on the two transition days, dropping the interpolated spring units and expanding the averaged autumn units to their two occurrences.

Figure 2 summarizes the empirical behavior of the post-transition target series in the available descriptive sample from 1 October 2025 to 31 July 2026. Prices are volatile and asymmetric, with realized values ranging from -499.99 to 747.10 EUR/MWh and 5.8% of observed MTUs taking negative values. The average intraday profile shows lower prices around midday and higher prices during morning and evening demand periods, which motivates the use of price lags, calendar variables, weather information, load, and renewable generation.

3.2.2 Input Data Sources and Derived Covariates

The market-price data consist of realized DE-LU Single Day-Ahead Coupling (SDAC) auction prices from the ENTSO-E Transparency Platform [17]. They define the price target and provide lagged price features that capture short-term dependence and recurring price patterns in electricity price forecasting [1, 2]. Prices from the Energy Exchange Austria (EXAA) are treated as an optional late market signal, because prices from this earlier auction carry predictive information for other European day-ahead markets [29]. The EXAA auction closes at 10:15 [31], and the resulting DE-LU day-ahead prices are published on the ENTSO-E Transparency Platform around

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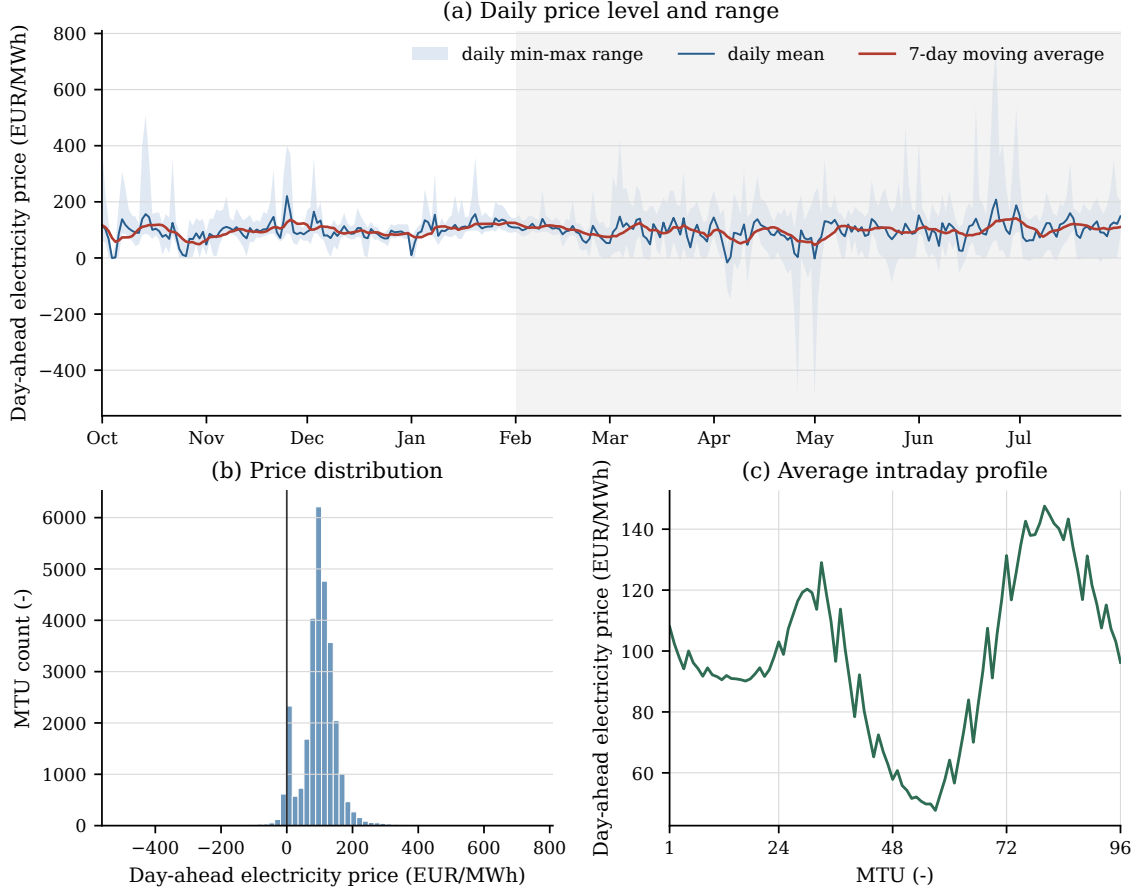


Figure 2: Empirical characteristics of the DE-LU day-ahead price target in the available post-transition sample from 1 October 2025 to 31 July 2026. The top panel shows daily mean prices together with the daily minimum-maximum range and a seven-day moving average. The shaded area marks the fixed out-of-sample test period from 1 February 2026 to 31 July 2026. The lower panels show the empirical price distribution and the average intraday price profile across MTUs.

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11:15 [17, 15]. Neighboring SDAC prices are excluded because they are released only after the 12:00 gate closure [3].

For the cutoff experiment, the price model also uses German control-reserve auction results from Regelleistung.net at every cutoff after their publication [32]. Control reserve is the balancing capacity that transmission system operators (TSOs) procure ahead of delivery to keep grid frequency stable, split by activation speed into frequency containment reserve (FCR), automatic frequency restoration reserve (aFRR), and manual frequency restoration reserve (mFRR), and cleared in daily capacity auctions on the day before delivery. The covariates summarize the published outcomes of these auctions. They are included because capacity committed to control reserve is withheld from the energy market and reserve prices reflect the expected scarcity of flexible capacity. The auction outcomes are therefore expected to carry an early signal of system tightness, which in a fundamental view shapes the day-ahead price through the conventional supply stack [5]. Together with the EXAA price signal, these reserve-market covariates are early market signals rather than explicit load or renewable-generation representations, so they are held out of RQ1 and RQ2, which study the OBTF load and renewable-generation forecasts and their price impact without early market information. They enter only in RQ3, where their staggered availability across the morning is part of the information-accrual design.

Load and renewable-generation data are also obtained from ENTSO-E [17]. For load, the pipeline uses lagged actual load, morning actual load observed before the forecast creation time, and the public day-ahead load forecast once it has been published before gate closure. For renewables, realized solar generation and realized onshore and offshore wind generation define the targets for the solar and wind forecasting models. The ENTSO-E day-ahead solar and wind generation forecasts serve as benchmarks for RQ1 but not as price model inputs, because they are not publicly observable before gate closure. The publication times of all inputs and their regulatory basis are documented in Section 3.3. Load, solar, and wind are economically relevant because demand and renewable feed-in shift the short-run supply-demand balance and thus the merit order [7].

Weather forecasts provide the main exogenous inputs for the OBTF forecasts of load, solar, and wind generation. The primary weather model is ICON-D2, the high-resolution short-range configuration of ICON that the DWD runs for Germany and its neighboring regions. Its forecasts are obtained both from DWD Open Data and through the Open-Meteo API [24, 25, 26]. A new run starts every three hours. Each run is named after its start time in Coordinated Universal Time (UTC), so the 06 UTC run starts at 08:00 local summer time. A run is complete about 80 minutes after its start, which makes the 06 UTC run available at about 09:20 local time in summer and one hour earlier in winter. All models rely on this single weather model. The OBTF wind model additionally uses its hub-height wind fields from Open-Meteo, described in Section 3.4.4. The weather variables include temperature, humidity, wind conditions, solar radiation, cloud cover, precipitation, snow, and atmospheric pressure. These variables are physically linked to load through heating and cooling demand [21], to photovoltaic generation through radiation and cloud cover [22], and to wind generation through wind speed and wind direction [23]. Installed renewable capacity from the MaStR connects the weather fields to the generation fleet [28], and gridded population data from Eurostat connect the weather fields to the demand side for load-oriented aggregation [27]. Calendar covariates comprise weekday indicators and a German public-holiday flag taken from

public-holiday calendars [33].

3.3 Operational Information Set

All model inputs are restricted to information that would have been available at the forecast creation time. For a local forecast creation time t on day $d - 1$, the admissible information set is

$$\mathcal{F}_{d-1}^t = \{Z_s \mid \text{avail}(Z_s) \leq \tau_{d-1}^t\},$$

where Z_s is an observation, forecast, or derived feature with reference time s , $\text{avail}(Z_s)$ is the time at which it becomes available to the pipeline, and τ_{d-1}^t is the clock time t on day $d - 1$. This rule is a leakage constraint. It does not imply that every model uses every available input.

In RQ1 and RQ2, forecasts are created with the latest feasible pre-gate-closure information set $\mathcal{F}_{d-1}^{12:00}$, with computation and submission completed slightly before 12:00. RQ3 varies the forecast creation time over the following times on day $d - 1$:

$$\mathcal{T}^{\text{cut}} = \{07:00, 08:00, 09:00, 10:00, 11:00, 12:00\}.$$

At each cutoff, every input block enters only once it is published, and the derived forecasts are rebuilt from the cutoff-specific information set. Figure 3 summarizes the timing of the main data blocks, Table 4 lists the inputs available at each cutoff, and Appendix A1 documents the sources behind the availability assumptions.

The availability constraint abstracts from computation time, treating t as a pure information cutoff and assuming the forecast is produced before gate closure. In a strict operational sense the admissible cutoff is earlier than t by the pipeline runtime, so a later cutoff, or a more expensive model, trades computation time for information. In the live deployment, the creation times t are the submission deadlines of the corresponding Energy-Arena challenges [18]. Each model run starts 20 minutes before its deadline and submits before it, so the operational information cutoff lies up to 20 minutes before t . Because the pipeline relies on lightweight boosted-tree models, this budget is short relative to the one-hour spacing of the cutoff grid, and the cutoff experiment isolates the value of information rather than a computation-time budget.

3.4 Forecasting Models

All learned forecasting tasks in this thesis use gradient-boosted regression trees [34]. The same model family is used for load, solar, wind, and price forecasting, which keeps the methodology consistent across tasks. Gradient-boosted trees are well suited to the heterogeneous tabular feature sets used here, on which they remain a strong and frequently superior alternative to deep-learning models [35, 36]. Boosted trees handle nonlinear relationships and interactions between weather, calendar, lagged values, and market variables without requiring these interactions to be specified manually [37, 34]. The implementation uses efficient histogram-based boosted-tree regressors from scikit-learn and LightGBM [38, 39]. Because both estimators belong to the same model family, the estimator choice is treated as an implementation detail and not as a separate research comparison.

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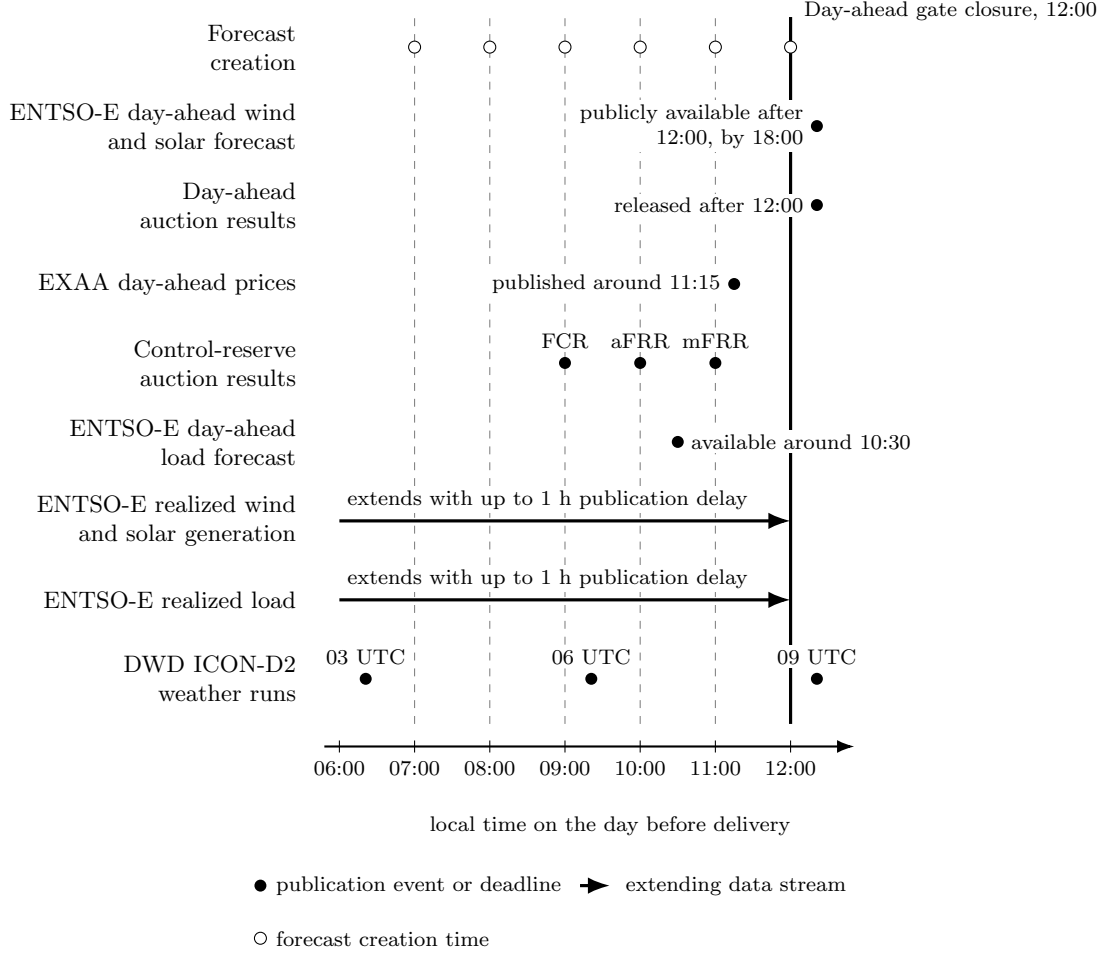


Figure 3: Operational overview of data availability, forecast creation times, and model runs on the day before delivery [16, 18, 31, 25, 26]. The dashed grid marks the forecast creation times in local time. Times are shown for summer time. Events to the right of 12:00 are not drawn to scale.

3.4.1 Weather Representations

All models use the DWD ICON-D2 forecast as their weather input (Section 3.2). ICON-D2 is a gridded numerical weather prediction that reports several weather variables at many grid points across the bidding zone, mostly at hourly resolution and for solar radiation at quarter-hourly resolution. Used directly, the number of raw grid features would far exceed the at most a few hundred daily training observations in the rolling training window, a regime in which boosted trees tend to overfit noise rather than learn the weather-to-target relationship. Because neighboring grid points are strongly correlated, most of this dimensionality is redundant. Each field is therefore converted into a fixed set of model features before it enters the load, solar, wind, and price models: the grid points are grouped into a small number of spatial clusters of nearby locations, and each cluster is summarized by a weighted spatial average of every weather variable. This limits the feature count and the resulting overfitting while retaining the weather signal relevant for price and generation. Clustering is preferred to a generic projection such as principal components [37] because it preserves the physical and spatial meaning of each feature and, unlike orthogonal components, lets the aggregation be weighted by where the target actually forms: population where demand matters and installed capacity where generation matters. Spatial aggregation of gridded weather into lower-dimensional features is also established practice in comparable operational forecasting pipelines [15]. For the OBTF load, solar, and wind models, the thesis does not study alternative clustering designs in a separate RQ, but describes the final spatial representations used in the pipeline. For the raw weather variables used directly by the price model, the spatial representation is instead selected on the development period, so that the comparison of the RQ2 base forecast $\hat{P}_{d,q|d-1}^{\text{base}}$ with the forecasts using OBTF inputs isolates the effect of the OBTF forecasts rather than that of a suboptimal weather aggregation. Starting from the five-cluster aggregation of the operational price forecasting pipeline of Eiser and Kleinebrahm [15], the number of clusters is varied and the development-period forecast error is minimized, which yields a two-cluster aggregation. Finer spatial resolutions do not improve development-period accuracy and raise the daily training cost steeply, as reported in Appendix A3. This selection remains a baseline design choice rather than a separate RQ.

For load, weather variables are summarized into DE-LU-level features on a 25-cluster weather grid. The spatial means are population-weighted, because demand depends on the weather experienced by consumers. Since load is concentrated where people live, an unweighted spatial mean would over-represent sparsely populated areas that contribute little demand, whereas a population-weighted mean better reflects the weather in the densely populated areas that drive most demand. Population-weighted temperature and the derived degree-day variables are long-established predictors in load forecasting [21]. The weighting is applied explicitly rather than left for the model to learn implicitly, because estimating the per-cluster weights from a short rolling training window would be data-inefficient and prone to overfitting, whereas the population-demand link is a reliable prior grounded in the load-forecasting literature. For solar generation, weather features are aligned with the distribution of installed photovoltaic capacity from the MaStR, using 25 clusters per TSO proxy area. A proxy area approximates a control area by the federal states it mainly covers, and installed capacity is assigned to it through the federal state recorded in the MaStR.

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For wind generation, weather features are aligned with installed wind capacity through a 100-cluster representation that includes offshore sites. This keeps the physical link between weather and the corresponding target. The cluster counts are chosen on the development period to balance the bias-variance trade-off rather than studied in a separate RQ: coarser grids oversmooth the field and underfit the capacity or demand distribution, while finer grids add correlated, redundant features that the data-limited models overfit. Solar and wind use finer grids than load because generation is concentrated at specific sites and spatially heterogeneous, whereas demand is smooth and is represented at the coarser DE-LU level. Wind in particular responds to local wind speed through a nonlinear power curve [40] and is concentrated at specific onshore and offshore sites. The resulting spatial representations are illustrated in Figure 4.

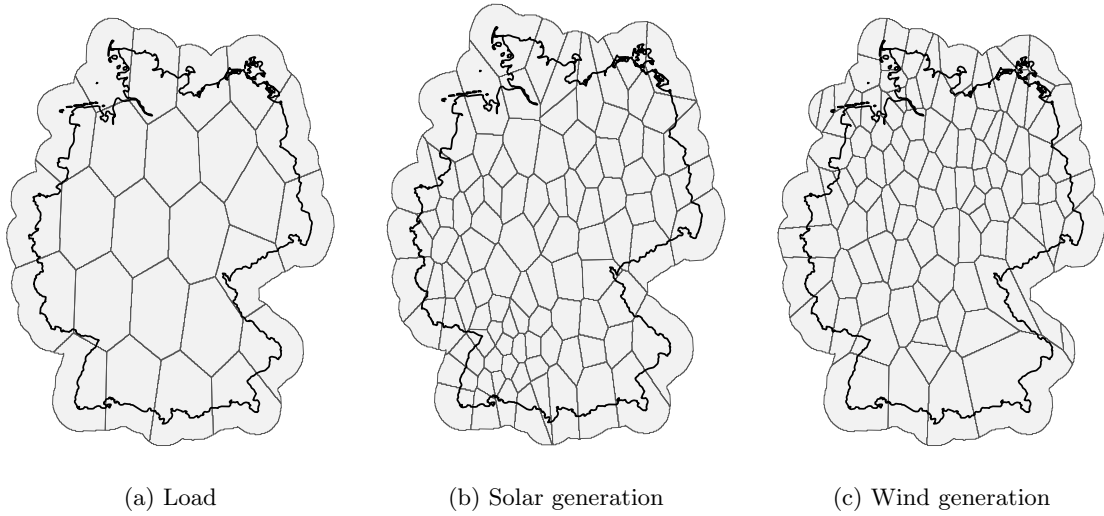


Figure 4: Spatial weather representations of the forecasting models. Panel (a) shows the 25-cluster weather grid used to construct population-weighted load-weather features. Panel (b) shows the solar representation with 25 clusters per TSO proxy area. Panel (c) shows the 100-cluster wind representation including offshore sites.

3.4.2 OBTF Load Model

The OBTF load model predicts realized DE-LU load $L_{d,q}$ for delivery day d and MTU q , producing a complete 96-MTU load curve for the delivery day. The resulting OBTF load forecast is denoted $\hat{L}_{d,q|d-1}$. The model is re-estimated in a rolling scheme with a boosted-tree regressor and uses calendar and holiday variables, lagged realized load, actual load observed during the morning before forecast creation, weather features, and, when available, the public ENTSO-E load forecast. For training, realized load of day $d - 1$ is used only up to the last quarter-hour published before forecast creation.

The weather block starts from cluster-level forecasts of temperature, dew point, surface pressure, precipitation, snow depth, cloud cover, solar radiation, and wind speed. From temperature and dew point, the pipeline derives relative humidity, vapor-pressure deficit, and heating and cooling degree variables. Rather than using the 25 cluster series individually, the OBTF load model summarizes

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them into a compact set of DE-LU-wide features. Population-weighted means describe the average weather exposure of consumers, while daily summaries capture the mean, minimum, maximum, and range of selected weather variables over the delivery day. Spatial weather quantiles describe the distribution across clusters at a given MTU, for example the lower, median, and upper part of the temperature or cloud-cover distribution. They are descriptive spatial quantiles, not probabilistic forecast quantiles, and are complemented by cross-cluster spread measures such as ranges and standard deviations. These features describe whether a weather extreme is confined to one part of the zone or spread across all of it, which affects aggregate demand beyond what the population-weighted mean captures.

The model has two operational modes. Before the public load forecast becomes available, it predicts load directly from the admissible input data. Once the public load forecast is available, the final model uses a forecast-correction specification. Let $L_{d,q|d-1}^{\text{ENTSOE}}$ denote the public ENTSO-E load forecast, with the correction target defined as the forecast deviation

$$\Delta_{d,q}^L = L_{d,q} - L_{d,q|d-1}^{\text{ENTSOE}},$$

and adds the predicted correction back to the public forecast,

$$\hat{L}_{d,q|d-1} = L_{d,q|d-1}^{\text{ENTSOE}} + \hat{\Delta}_{d,q|d-1}^L.$$

This treats the ENTSO-E forecast as an admissible baseline and uses the model to correct remaining errors with the available load, calendar, and weather information. The public forecast is produced by the system operators with undisclosed models and inputs [16, 17], which cannot be reconstructed from the open data available here. Correcting it lets the model add whatever information the public forecast does not already reflect, namely its systematic errors, information that arrived after it was issued, such as newer weather runs and the morning’s realized load, and information that was available but is not exploited by the undisclosed model. Predicting this correction rather than the load level is also favorable in bias-variance terms: the correction is a small and comparatively stationary target with far less variance than the full load curve, so a model trained on only a few hundred rolling days estimates it more reliably than it could estimate load from scratch. The direct mode is relevant for early forecast creation times in RQ3, while the correction mode is used whenever the public load forecast is available.

3.4.3 OBTF Solar Model

The OBTF solar model predicts realized solar generation $S_{d,q}$ for all 96 MTUs of the delivery day. The resulting OBTF solar forecast is denoted $\hat{S}_{d,q|d-1}$. In the empirical analysis, the model is implemented with histogram-based boosted trees and estimated on a rolling training window. It is trained separately for the four German control areas of 50Hertz, Amprion, TenneT, and TransnetBW, because realized solar generation is published per control area and the bidding-zone value is essentially their sum. In the test period, the four areas account for 99.4% of realized DE-LU solar generation, with the remainder from Luxembourg. Summing the four area forecasts to the aggregate OBTF solar forecast therefore mirrors how the realized value is composed.

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The inputs are calendar variables, installed photovoltaic capacity, the position of the sun, and weather. Realized generation is not used as a lagged input. It enters only as the training target and through the bias correction described below. The weather is aggregated over 25 clusters per TSO proxy area (Section 3.4.1) and weighted by installed photovoltaic capacity. It includes irradiance, temperature, precipitation, snow, and cloud cover, together with spread measures that distinguish uniformly cloudy areas from areas where conditions differ across locations.

The model also uses a physics-based solar baseline, the expected photovoltaic generation computed from the forecast irradiance and temperature, the position of the sun, and the installed capacity [22]. The boosted-tree model then predicts only the deviation from this baseline, which keeps the forecast physically plausible. Training days on which realized generation falls far below the baseline under clear-sky conditions, a pattern more consistent with data errors or curtailment than with weather, are excluded. Predictions are finally adjusted with a rolling bias correction that subtracts the average error of recent days, applied per control area and to the DE-LU total, which also covers the small Luxembourg share.

3.4.4 OBTF Wind Model

The OBTF wind model predicts realized wind generation $W_{d,q}$, including both onshore and offshore generation, for all 96 MTUs of the delivery day. The resulting OBTF wind forecast is denoted $\hat{W}_{d,q|d-1}$. In the empirical analysis, the model is implemented with histogram-based boosted trees and estimated on a rolling training window. Onshore and offshore generation are forecast separately, because they differ in their weather drivers and spatial concentration and are published as separate production types [17]. The two forecasts are then summed to the aggregate OBTF wind forecast used by the price model.

The inputs are calendar variables, installed wind capacity, and weather. As for solar, realized generation enters only as the training target and through the bias correction. The weather is aggregated over 100 clusters including offshore sites (Section 3.4.1), grouped into an offshore region and five onshore regions, and weighted by installed wind capacity. It includes wind speed and direction, temperature, and precipitation. Because wind generation rises nonlinearly with wind speed, the model also uses transformed wind speeds that approximate a turbine power curve [40]. From the same ICON-D2 run, accessed through Open-Meteo [26], it additionally uses wind speeds at typical turbine hub heights between 80 and 180 m, because generation is especially sensitive to the wind at hub height [40, 23]. Predictions are corrected with the same rolling bias correction as for solar.

3.4.5 Price Model

The price model predicts $P_{d,q}$ with boosted trees. Before fitting, the price target is robustly standardized and passed through an inverse hyperbolic sine transform [41], which stabilizes the variance of the volatile, spike-prone, and occasionally negative price series, and predictions are transformed back to price levels for evaluation. As is common in day-ahead price forecasting, a separate model is fitted for each MTU, because the price level and its drivers differ strongly across the day [2]. In RQ2, the base price forecast $\hat{P}_{d,q|d-1}^{\text{base}}$ is compared with the price forecast with OBTF

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load and renewable-generation forecasts $\hat{P}_{d,q|d-1}^{\text{gen}}$. The base price forecast uses price lags, calendar variables, weather variables, and the published ENTSO-E day-ahead load forecast, and excludes early cross-market price signals. The price forecast with OBTF load and renewable-generation forecasts uses the same variables but replaces the ENTSO-E load forecast with the OBTF load forecast $\hat{L}_{d,q|d-1}$ and adds the OBTF combined renewable-generation forecast $\hat{G}_{d,q|d-1}^{\text{ren}}$ and the OBTF residual-load forecast $\widehat{RL}_{d,q|d-1}$. Because these inputs are themselves model forecasts rather than realized values, they enter the price model as forecasted inputs and carry their own forecast error [42]. Solar and wind enter the price model as a single combined forecast rather than as two separate inputs, because price formation responds to the total renewable feed-in that lowers residual load rather than to the split between the two sources, and a single input keeps the data-limited price model parsimonious. Both are still forecast separately by their own models. Residual load denotes load net of renewable generation and is also referred to as residual demand or net load in the literature [6]. It is included as an explicit input in addition to load and combined renewable generation because it directly represents the demand that must be met by dispatchable plants, which is the quantity that sets the marginal price in the merit order [7]. Providing residual load explicitly allows a single tree split to represent this supply-demand balance, which the boosted-tree model would otherwise have to recover from load and renewable generation through feature interactions. This comparison isolates the value of $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$ because the remaining price lags, calendar variables, and weather variables are kept the same. A weather-free price forecast with OBTF load and renewable-generation forecasts, $\hat{P}_{d,q|d-1}^{\text{gen},-\text{wea}}$, keeps these forecasts but omits the raw weather features. Because OBTF forecasts are themselves derived from weather, this variant tests whether they provide a representation of the weather-driven system state that can substitute for the raw weather inputs. The cluster-level weather representations are not passed on to the price model, but are used only inside the OBTF models. These models first generate final forecasts for load, solar, and wind generation for each MTU, and the price model receives only the three DE-LU totals $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$. Intermediate outputs of the OBTF models, such as the separate OBTF solar and wind forecasts before they are combined or any regional and cluster-level series, are not passed on. The RQ3 timing experiment then changes the admissible input set according to the forecast creation time. Additional inputs enter only after their publication. In particular, reserve-market covariates are included only after the corresponding Regelleistung.net auction-result blocks are available, and EXAA prices enter only at the 12:00 cutoff.

Two design choices of the price model are fixed on the development period before the test period is used. The raw weather variables used by the price model are aggregated to two spatial clusters, as described in Section 3.4.1, and the rolling training window is set to 70 days, the shortest length beyond which development-period accuracy no longer improves. The window is short because price relationships shift quickly with fuel prices, renewable capacity, and season, so recent days are more informative than older ones. These choices are made with the same boosted-tree setup for $\hat{P}_{d,q|d-1}^{\text{base}}$ and $\hat{P}_{d,q|d-1}^{\text{gen}}$, so that the RQ2 comparison is not confounded by a weaker base price forecast. Appendix A3 reports both selection sweeps, including the steep growth of the daily training cost with the number of weather clusters.

Two simple benchmark forecasts for the price forecast are included in RQ2. The previous-day

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naive benchmark uses the realized price curve of delivery day $d - 1$ as the forecast for day d . The previous-week naive benchmark uses the realized price curve of delivery day $d - 7$. Both benchmarks are admissible before the 12:00 gate closure and provide an easily interpretable baseline for the model-based price forecasts. Table 2 summarizes the price forecast variants used in RQ2 and RQ3.

Table 2: Price forecast variants used in RQ2 and RQ3.

Variant	Description
N_{d-1}	Previous-day naive price forecast using the realized price curve of delivery day $d - 1$.
N_{d-7}	Previous-week naive price forecast using the realized price curve of delivery day $d - 7$.
$\hat{P}_{d,q d-1}^{\text{base}}$	Base price forecast from the boosted-tree model using the ENTSO-E load forecast but without OBTF load, renewable-generation, and residual-load forecasts.
$\hat{P}_{d,q d-1}^{\text{gen}}$	Price forecast with OBTF load and renewable-generation forecasts, using the same boosted-tree model structure as $\hat{P}_{d,q d-1}^{\text{base}}$, replacing the ENTSO-E load forecast with $\hat{L}_{d,q d-1}$ and adding $\hat{G}_{d,q d-1}^{\text{ren}}$ and $\hat{R}_{d,q d-1}$ as inputs.
$\hat{P}_{d,q d-1}^{\text{gen}, -\text{wea}}$	Weather-free price forecast with OBTF load and renewable-generation forecasts. It keeps these forecasts but omits the raw weather features.
$\hat{P}_{d,q d-1}^t$	Cutoff-specific RQ3 price forecast for $t \in \mathcal{T}^{\text{cut}}$, using the weather, morning actuals, OBTF load and renewable-generation forecasts, reserve-market features, public ENTSO-E load forecast, and EXAA price signal that are admissible at cutoff t .

3.5 Experimental Design

All experiments share the same temporal setup. Models are re-estimated daily in a rolling scheme: for each delivery day, each model is trained on a preceding rolling window of fixed length and then produces the forecast for that day. A fixed-length rolling window, rather than an expanding one, is used because the DE-LU system is non-stationary: installed capacity, demand, and price behavior drift over time, so recent history is more representative of the delivery day, and bounding the window keeps the model adaptive and the daily training cost fixed. For the base price model, the development-period sweep shows no meaningful accuracy gain from an expanding window (Appendix A3.1). The post-transition sample is split into a development period from 1 October 2025 to 31 January 2026 and an out-of-sample test period from 1 February 2026 to 31 July 2026. The development period is used to fix all design choices, including feature sets, model hyperparameters, and the rolling-window length, while the test period is held out and used only for the reported results, so that no design decision depends on test-period information. The window length is selected per model on the development period, which for the price model yields 70 days. It is kept short enough that the price model is trained entirely on post-transition quarter-hourly prices for every

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test day, so no reported result relies on pre-transition prices, which are available only at hourly resolution. Such hourly prices enter only the training windows of the earliest development days, which reach before 1 October 2025. Evaluations that involve the OBTF wind forecast exclude 7 February 2026, for which no ICON-D2 run with wind fields is archived. The test period spans late winter, spring, and summer, which matters because load, solar, and wind generation follow different seasonal patterns and a single-season window would test them unevenly.

3.5.1 Research Design Matrix

Table 3 links the empirical design to the three RQs. The design covers the accuracy of OBTF forecasts, their effect on the price forecast, and changes in price forecast performance as richer information sets become available before gate closure. The weather representation and spatial aggregation of the OBTF load, solar, and wind models are fixed parts of the implemented pipeline rather than separate experimental factors. The weather aggregation used by the price model is the one exception, and it is selected on the development period to keep the base price forecast in RQ2 a fair control (Section 3.4.1).

Table 3: Research and hypothesis design matrix.

RQ/H	Empirical design	Outcome assessed
RQ1/H1	$\hat{L}_{d,q d-1}$, $\hat{S}_{d,q d-1}$, and $\hat{W}_{d,q d-1}$ are compared with $L_{d,q d-1}^{\text{ENTSOE}}$, $S_{d,q d-1}^{\text{ENTSOE}}$, and $W_{d,q d-1}^{\text{ENTSOE}}$. The realized targets are $L_{d,q}$, $S_{d,q}$, and $W_{d,q}$.	Forecast quality of the OBTF forecasts $\hat{L}$, $\hat{S}$, and $\hat{W}$ relative to ENTSO-E benchmarks
RQ2/H2	$\hat{P}_{d,q d-1}^{\text{gen}}$ is compared with $\hat{P}_{d,q d-1}^{\text{base}}$. The two forecasts differ only in whether $\hat{L}_{d,q d-1}$, $\hat{G}_{d,q d-1}^{\text{ren}}$, and $\widehat{R}\hat{L}_{d,q d-1}$ enter the price model.	Effect of the OBTF load and renewable-generation forecasts on price forecast accuracy
RQ3/H3	The forecasting pipeline is rerun for $t \in \mathcal{T}^{\text{cut}}$. Inputs are restricted to what is available at each cutoff, including the staggered arrival of reserve-market results and the late EXAA price signal.	Price forecast performance with richer information sets over time

3.5.2 RQ1: OBTF Forecasts versus ENTSO-E Forecasts

RQ1 is addressed by comparing $\hat{L}_{d,q|d-1}$, $\hat{S}_{d,q|d-1}$, and $\hat{W}_{d,q|d-1}$ with the corresponding ENTSO-E day-ahead forecasts. All forecasts are aligned to the same 96-MTU delivery-day grid and evaluated against realized load, solar, and wind generation after delivery. The comparison is deliberately not a same-information comparison: OBTF forecasts are restricted to the operational information cutoff of the price pipeline, while the ENTSO-E day-ahead solar and wind generation benchmark forecasts are not publicly observable before the 12:00 gate closure. This difference reflects the operational setting rather than a flaw of the design, because the OBTF forecasts have to be created before

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gate closure so that they can enter the price model, a constraint the ENTSO-E solar and wind benchmarks do not face. Figure 5 illustrates this comparison.

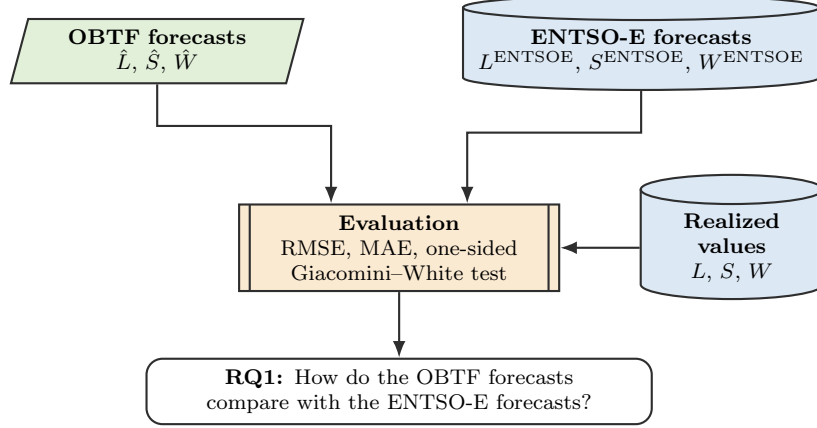


Figure 5: Design of RQ1. The OBTF load, solar, and wind forecasts and the ENTSO-E forecasts are each scored against the realized values on a matched panel. The OBTF forecasts use only information available before the 12:00 deadline. Notation as in Figure 1.

3.5.3 RQ2: Explicit Load and Renewable-Generation Forecasts in Price Forecasting

RQ2 is addressed by testing how an explicit representation of load and renewable-generation forecasts affects the price forecast. This is implemented by adding $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$ to the price model. The main comparison is between $\hat{P}_{d,q|d-1}^{\text{base}}$ and $\hat{P}_{d,q|d-1}^{\text{gen}}$. Both forecasts are produced with the same boosted-tree model class, the same forecast creation time, the same evaluation window, and the same price-lag, calendar, and raw weather inputs. They differ only in whether $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$ are included. Since OBTF forecasts are produced by the separate OBTF load, solar, and wind models with task-specific weather representations, this comparison measures the effect of the explicit load and renewable-generation forecast representation rather than the effect of an identical raw weather information set.

A secondary comparison uses the weather-free variant $\hat{P}_{d,q|d-1}^{\text{gen}, -\text{wea}}$, which keeps OBTF forecasts but omits the raw weather features. Because OBTF forecasts are derived from weather, comparing $\hat{P}_{d,q|d-1}^{\text{gen}, -\text{wea}}$ with $\hat{P}_{d,q|d-1}^{\text{base}}$ tests whether they can stand in for the raw weather features, while comparing it with $\hat{P}_{d,q|d-1}^{\text{gen}}$ shows whether the raw weather features still add information beyond OBTF forecasts.

The previous-day and previous-week naive price forecasts are included in this price forecast comparison. They are not used as benchmarks for the load, solar, and wind forecasts because RQ1 concerns the quality of OBTF forecasts relative to the corresponding ENTSO-E day-ahead forecasts. In RQ2, their role is to show whether the price models add value beyond simple price persistence. Figure 6 illustrates the RQ2 comparison together with the weather-free variant and the naive benchmarks.

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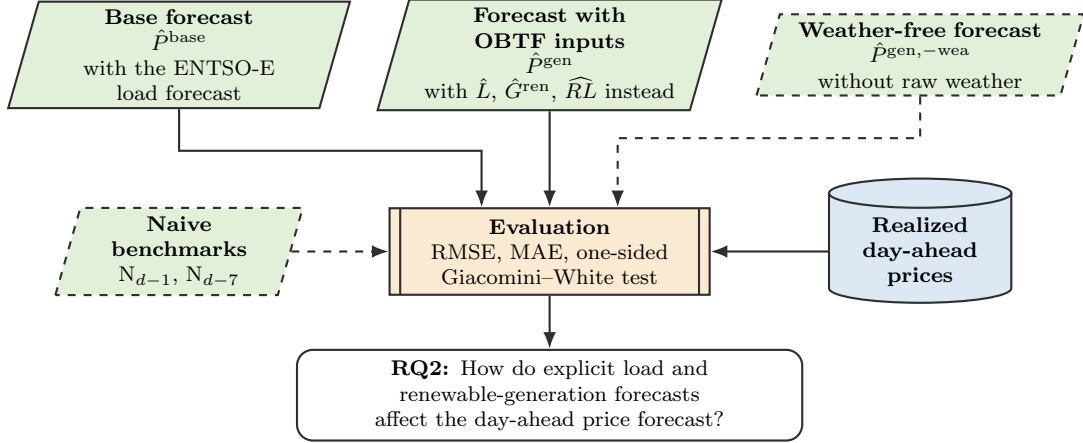


Figure 6: Design of RQ2. The base forecast uses price history, calendar features, weather, and the ENTSO-E load forecast, while $\hat{P}^{\text{gen}}$ replaces the latter with the OBTF load, renewable-generation, and residual-load forecasts. A weather-free variant and two naive forecasts serve as secondary comparisons. Notation as in Figure 1.

3.5.4 RQ3: Richer Information Sets over Time

RQ3 is addressed through a timing experiment that evaluates how electricity price forecast performance changes with richer information sets over time. The forecasting pipeline is rerun at the cutoff times $\mathcal{T}^{\text{cut}}$ from Section 3.3. At each cutoff, the OBTF load, solar, and wind models of RQ1 and RQ2 are rerun unchanged with only the inputs available before that time on day $d - 1$, so the OBTF forecasts entering the price model are themselves cutoff-specific. Realized load and generation enter only up to the last quarter-hour published at the start of the model run, allowing for a publication delay of up to one hour [16]. All cutoffs use the ICON-D2 06 UTC run, which is complete about 80 minutes after its start and is therefore admissible from the 10:00 cutoff onward in both winter and summer time. Operationally, the earlier cutoffs would use an earlier run (Table 4). In the backtest this is not possible, because the later 09 UTC run is complete only after 12:00 in summer and the earlier 00 and 03 UTC runs were not archived for the first months of the test period, so the earlier cutoffs also use the 06 UTC run (Section 5.5). Because all cutoffs share this weather run, moving later in the morning can improve the forecasts through more recent realized load and generation values, reserve-market auction results, and additional inputs with later publication times. For load, this also changes the admissible model setup: early cutoffs rely on the direct OBTF load model, while cutoffs after publication of the public ENTSO-E load forecast can use the forecast-correction mode from Section 3.4.2. Reserve-market results enter in stages as the corresponding product auctions become observable: FCR from 09:00, aFRR from 10:00, and mFRR from 11:00. The EXAA day-ahead price, published around 11:15, becomes admissible only at the 12:00 cutoff and is included there as a late price signal. Table 4 lists the admissible inputs at each cutoff. Figure 7 illustrates the timing experiment.

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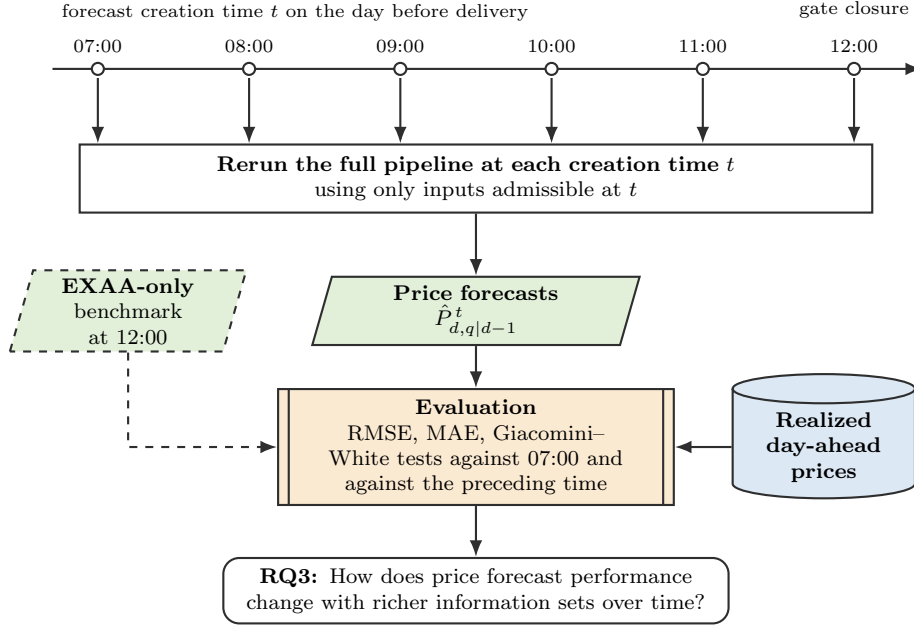


Figure 7: Design of RQ3. The full pipeline is rerun at six forecast creation times from 07:00 to the 12:00 gate closure, each time with only the information admissible at that time, and the resulting price forecasts are compared with each other and with a compact EXAA-only benchmark. Notation as in Figure 1.

Table 4: Admissible inputs at each forecast creation time on day $d - 1$. A dash denotes an input that is not yet admissible. ^aDue to data limitations, the backtest uses the 06 UTC run at every cutoff (Section 5.5).

Cutoff	Weather run		Realized data to	OBTF load model	Reserve products	EXAA
	Operation	Backtest ^a				
07:00	03 UTC	06 UTC	05:30	Direct	—	—
08:00	03 UTC	06 UTC	06:30	Direct	—	—
09:00	03 UTC	06 UTC	07:30	Direct	FCR	—
10:00	06 UTC	06 UTC	08:30	Direct	FCR, aFRR	—
11:00	06 UTC	06 UTC	09:30	Correction	FCR, aFRR, mFRR	—
12:00	06 UTC	06 UTC	10:30	Correction	FCR, aFRR, mFRR	✓

3.6 Evaluation Framework

3.6.1 Evaluation Panel and Error Metrics

The evaluation framework is shared across all RQs. Each comparison uses out-of-sample forecasts generated under the admissible information set for the corresponding forecast creation time, over the test period from 1 February 2026 to 31 July 2026 defined in Section 3.5. Models within the same RQ are evaluated on the same delivery days and MTUs. If a model is missing a forecast for a timestamp in the matched panel, the timestamp is removed for all models in that specific comparison and the number of observations is reported.

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Let $e_{d,q}^m = Y_{d,q} - \hat{Y}_{d,q|d-1}^m$ denote the forecast error of model m , and let $\mathcal{E}$ be the matched evaluation panel. All point models are estimated by minimizing squared error, so their forecasts target the conditional mean of the delivery-period outcome. The primary metric is therefore the root mean squared error (RMSE),

$$\text{RMSE}(m) = \sqrt{\frac{1}{|\mathcal{E}|} \sum_{(d,q) \in \mathcal{E}} (e_{d,q}^m)^2},$$

which rewards accurate estimation of that conditional mean and is the scoring metric of the Energy-Arena submissions [18]. The mean absolute error (MAE) $\text{MAE}(m) = |\mathcal{E}|^{-1} \sum_{(d,q) \in \mathcal{E}} |e_{d,q}^m|$ is reported alongside it for every comparison, because it is a standard accuracy metric in the electricity price forecasting literature and, being expressed in the target unit, aids interpretation and comparison with published work [2, 43]. Price metrics are reported in EUR/MWh. Load, solar, and wind metrics are reported in MW.

3.6.2 Paired Model Comparisons

Model comparisons are reported as relative RMSE changes. For a reference forecast a and a challenger b ,

$$\Delta \text{RMSE}_{a,b} = \frac{\text{RMSE}(b) - \text{RMSE}(a)}{\text{RMSE}(a)},$$

so a negative value means that b has a lower error than a . When statistical significance is assessed, the thesis uses the Giacomini-White (GW) conditional predictive ability test [44] on the squared-error loss, consistent with the squared-error training objective and the RMSE metric. Because the 96 quarter-hourly prices of a delivery day are forecast jointly, the test is applied to the daily mean loss differential $\bar{\delta}_d$ across the 96 MTUs. The test uses two instruments, namely a constant and the previous day's differential $\bar{\delta}_{d-1}$, so the test statistic is asymptotically χ^2 -distributed with two degrees of freedom. The test is used one-sided, in the direction of the forecast with the lower average loss. The test provides supporting evidence, while the main interpretation is based on the size, direction, and consistency of the out-of-sample error differences. When several tests address the same question, the chance that at least one of them appears significant by coincidence grows with their number. Within each such family, namely each column of the RQ3 cutoff results and the three pre-EXAA cutoffs of the reserve ablation, the p-values are therefore additionally checked with the Holm procedure [45]. It orders the m p-values of a family from smallest to largest, compares the smallest with $5\%/m$, the next with $5\%/(m-1)$, and so on, and stops at the first one that exceeds its threshold. This keeps the probability of any false significant result within the family at 5%.

3.6.3 Feature Importance

As a diagnostic, the boosted-tree price model is analyzed with SHapley Additive exPlanations (SHAP) values [46, 47]. SHAP values attribute each individual prediction to its input features and are aggregated by summation within the feature groups that make up the price model: price lags, calendar and holiday variables, weather variables, the EXAA price and reserve-market auction

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results where admissible, and the OBTF forecasts $\hat{L}_{d,q|d-1}$, $\hat{G}_{d,q|d-1}^{\text{ren}}$, and $\widehat{RL}_{d,q|d-1}$. Because SHAP values are signed, they also show the direction of each feature's effect, that is, whether it raises or lowers the forecast, which summary plots across all predictions make visible [46, 47], as in Figure A2. This analysis helps interpret whether the OBTF forecasts are actually used by the price model.

4 Results

4.1 RQ1: OBTF Forecasts versus ENTSO-E Forecasts

Table 5 compares OBTF forecasts with the corresponding ENTSO-E day-ahead forecasts, each evaluated against the same realized value. For load, the OBTF forecast is the forecast-correction model of Section 3.4.2, created at the final pre-gate-closure run before 12:00, so the comparison is against the uncorrected ENTSO-E load forecast that this model takes as its baseline. The direct OBTF load model, which uses no ENTSO-E forecast, is not part of the fixed comparison in Table 5, but is benchmarked against ENTSO-E separately below and enters RQ3 at cutoff times before the public ENTSO-E load forecast is available. For solar and wind, the ENTSO-E benchmark forecast is missing for a small number of timestamps, which are dropped from the matched panel.

Table 5: Accuracy of OBTF load, solar, and wind forecasts compared with ENTSO-E forecasts.

Target	n	RMSE		Δ RMSE ^a	MAE		p_{GW} ^b
		OBTF	ENTSO-E		OBTF	ENTSO-E	
Load	17,376	1,727	2,697	-36.0%	1,350	2,140	< 0.001
Solar gen.	17,280	2,301	1,370	+68.0%	1,109	747	< 0.001
Wind gen.	17,184	3,119	2,248	+38.8%	2,092	1,450	< 0.001

^a Δ RMSE is computed relative to the corresponding ENTSO-E forecast. Negative values indicate lower errors of the OBTF forecast. All error values are in MW. The column n gives the number of matched observations used for the comparison. The evaluation window is 1 February 2026 to 31 July 2026. The renewable-generation comparisons have fewer observations because the ENTSO-E solar and wind benchmark forecasts are missing for some timestamps, and the wind comparison excludes 7 February 2026, for which no ICON-D2 run with wind fields is archived.

^b One-sided GW p-value for the forecast with the lower RMSE against the alternative forecast, using paired squared-error loss differences.

The load result supports the first part of H1. The OBTF load forecast reduces the RMSE from 2,697 MW to 1,727 MW, a 36.0% error reduction relative to the ENTSO-E load forecast, with a comparable reduction in MAE. This pattern is consistent with the model design, because the OBTF load forecast used here corrects the public ENTSO-E forecast instead of replacing it with an independent direct forecast. It can therefore combine the public forecast with recent morning load and weather information available before gate closure. Because the two forecasts do not share an information cutoff, this is an operational rather than a like-for-like modeling comparison: the ENTSO-E forecast is produced by an undisclosed model and issued at a generally earlier, unspecified time, so the 36.0% reduction measures the operational value of the additional pre-gate-closure information the correction exploits rather than a difference in modeling skill alone. A stricter comparison nonetheless isolates a genuine modeling component: the direct OBTF load model, which uses only open data and does not use the ENTSO-E forecast at all, still reduces RMSE by 17.3% relative to ENTSO-E when it is formed at the 10:30 cutoff at which the public forecast is published, a difference that is statistically significant (GW p-value of 0.003). Correcting the public forecast then extends this advantage to 36.0%. Within this operational setting, the OBTF load forecast is not only competitive with the ENTSO-E forecast, but substantially more accurate in this test window. The GW test supports this ranking with a p-value below 0.001.

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The renewable-generation comparison shows the opposite pattern. The OBTF solar forecast has an RMSE of 2,301 MW compared with 1,370 MW for the ENTSO-E solar forecast. The OBTF wind forecast has an RMSE of 3,119 MW compared with 2,248 MW for the ENTSO-E wind forecast. The ENTSO-E renewable-generation forecasts therefore outperform the OBTF solar and wind forecasts in this benchmark comparison. The GW test also supports the ENTSO-E ranking for both renewable-generation targets. For solar and wind, however, the ENTSO-E forecasts are not admissible inputs for the operational price forecast studied in this thesis. The ENTSO-E forecasts used here are not publicly observable before the 12:00 day-ahead gate closure, while the OBTF solar and wind forecasts are restricted to information that is available before gate closure. The higher errors of the OBTF solar and wind forecasts therefore do not rule out their relevance for the price model.

Overall, H1 is supported. The OBTF load forecast is not only competitive with but more accurate than the ENTSO-E forecast, and the OBTF solar and wind forecasts have higher errors than the ENTSO-E benchmarks, as expected. The renewable gap is substantial, however, with an RMSE that is 68.0% higher for solar and 38.8% higher for wind. The result therefore establishes the empirical trade-off for RQ2: the OBTF solar and wind forecasts are less accurate than their ENTSO-E benchmarks, but they are admissible inputs for the price forecast. Whether this operationally available representation still improves the price forecast is tested in RQ2.

4.2 RQ2: Explicit Load and Renewable-Generation Forecasts in Price Forecasting

RQ2 tests how an explicit representation of load and renewable-generation forecasts affects the day-ahead price forecast. This representation consists of the OBTF load forecast, the OBTF combined renewable-generation forecast, and the OBTF residual-load forecast. Table 6 reports the out-of-sample accuracy of the two naive benchmarks and the three price forecasts from the boosted-tree price model, all evaluated on the same matched panel of delivery days and MTUs.

Table 6: Out-of-sample price forecast accuracy of the naive benchmarks and the price forecasts with and without OBTF load and renewable-generation forecasts.

Forecast	n	RMSE	ΔRMSE^a	MAE	p_{GW}^b
N_{d-1}	17,184	49.69	+40.0%	30.63	< 0.001
N_{d-7}	17,184	60.31	+69.9%	38.18	< 0.001
$\hat{P}^{\text{base}}$	17,184	35.50	—	20.82	—
$\hat{P}^{\text{gen}}$	17,184	31.54	-11.2%	18.05	< 0.001
$\hat{P}^{\text{gen}, -\text{wea}}$	17,184	31.79	-10.5%	18.18	< 0.001

^a ΔRMSE is computed relative to $\hat{P}^{\text{base}}$. Negative values indicate lower errors than the base price forecast. All error values are in EUR/MWh. The column n gives the number of matched observations. The evaluation window is 1 February 2026 to 31 July 2026, with 7 February 2026 (no archived wind input) and 19 June 2026 (weather input unavailable) removed.

^b One-sided GW p-value between the row forecast and $\hat{P}^{\text{base}}$, for the forecast with the lower RMSE of the two, using paired squared-error loss differences.

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Explicitly representing load and renewable generation improves the price forecast. The price forecast with OBTF load and renewable-generation forecasts, $\hat{P}^{\text{gen}}$, reduces the RMSE from 35.50 EUR/MWh for the base forecast $\hat{P}^{\text{base}}$ to 31.54 EUR/MWh, a reduction of 11.2%, with a comparable reduction in MAE. The GW test rejects equal predictive accuracy in favor of $\hat{P}^{\text{gen}}$, with a p-value below 0.001. H2 is therefore supported: the explicit load and renewable-generation forecast representation carries information that $\hat{P}^{\text{base}}$ does not already recover from price lags, calendar variables, raw weather, and the ENTSO-E load forecast.

The weather-free variant $\hat{P}^{\text{gen}, -\text{wea}}$ isolates where this information comes from. It keeps the OBTF forecasts but drops the raw weather features. Its RMSE of 31.79 EUR/MWh is statistically indistinguishable from $\hat{P}^{\text{gen}}$, with a GW p-value of 0.20, and it still improves on $\hat{P}^{\text{base}}$ by essentially the same margin, with a p-value below 0.001. The raw weather features therefore add no measurable accuracy once OBTF forecasts are included. Because the OBTF load, solar, and wind forecasts are themselves derived from weather, this is consistent with them carrying the price-relevant part of the weather information available to the model.

Both naive benchmarks are far less accurate than the model-based forecasts. The previous-day benchmark N_{d-1} has an RMSE of 49.69 EUR/MWh and the previous-week benchmark N_{d-7} an RMSE of 60.31 EUR/MWh, against 35.50 EUR/MWh for $\hat{P}^{\text{base}}$ and 31.54 EUR/MWh for $\hat{P}^{\text{gen}}$. All three model-based price forecasts ($\hat{P}^{\text{base}}$, $\hat{P}^{\text{gen}}$, and $\hat{P}^{\text{gen}, -\text{wea}}$) outperform both benchmarks with GW p-values below 0.001, and N_{d-1} in turn outperforms N_{d-7} . The model-based price forecasts therefore add substantial value beyond simple price persistence.

Taken together with RQ1, this resolves the trade-off identified there. The OBTF solar and wind forecasts are less accurate than the ENTSO-E benchmarks, but those benchmarks are not publicly observable before gate closure and are not admissible for the price forecast. The OBTF forecasts, which are available before gate closure, still let $\hat{P}^{\text{gen}}$ improve significantly on $\hat{P}^{\text{base}}$, which already uses price lags, calendar variables, raw weather, and the ENTSO-E load forecast.

4.3 RQ3: Richer Information Sets over Time

RQ3 evaluates how the operational day-ahead price forecast changes as the information set expands during the morning of day $d - 1$. All models predict the DE-LU day-ahead price at quarter-hourly resolution and are evaluated on the same test period as RQ2. The cutoff-specific OBTF load, solar, and wind forecasts are generated separately for each forecast creation time, so that the price model receives only information that would have been available at the corresponding cutoff. The 07:00, 08:00, 09:00, and 10:00 specifications use the direct OBTF load forecast, while the 11:00 and 12:00 specifications use the correction of the public ENTSO-E day-ahead load forecast. Reserve-market features enter only after the assumed publication cutoff of the corresponding product: FCR from 09:00, FCR and aFRR from 10:00, and FCR, aFRR, and mFRR from 11:00 onward. EXAA prices enter only at the 12:00 cutoff.

Table 7 and Figure 8 report the cutoff-grid results. Forecast accuracy improves over the morning, but gradually and without a statistically significant step before 12:00. The 07:00 model reaches an RMSE of 32.63 EUR/MWh. The 08:00 and 09:00 models are close to it, at 32.83 and 32.39 EUR/MWh. Neither improves significantly on the 07:00 baseline, with GW p-values of 1.00

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and 0.42, and neither improves significantly on the cutoff one hour earlier.

From 10:00 onward the error declines more clearly. The 10:00 model, which adds the aFRR results to the FCR results, reaches an RMSE of 31.78 EUR/MWh. The 11:00 model, which additionally includes the mFRR results and switches to the load correction mode, reaches 30.81 EUR/MWh, 5.6% below the 07:00 baseline. Neither gain is statistically significant, however. Against the 07:00 baseline the p-values are 0.083 and 0.054, and none of the steps between consecutive cutoffs before 12:00 is significant. The direction of the 11:00 reduction is consistent with RQ1. Once the public ENTSO-E load forecast is published around 10:30, the OBTF load model switches from its direct forecast to the more accurate correction mode. A reserve-market ablation (Appendix A3.2) shows that the reserve-market results lower the error at every cutoff before 12:00, most at 10:00 with 3.4%. None of these reductions is significant after the Holm correction.

The largest improvement occurs once EXAA prices become available. The full 12:00 model, which combines residual load, OBTF renewable-generation forecasts, weather, all reserve products, and EXAA prices, reaches an RMSE of 27.58 EUR/MWh, 10.5% below the 11:00 forecast. This is the only step that significantly improves on its preceding cutoff, and it also significantly improves on the 07:00 baseline ($p < 0.001$), in both cases also after the Holm correction. The compact EXAA-only benchmark is slightly more accurate, with an RMSE of 27.13 EUR/MWh. This indicates that EXAA prices contain a very strong late-market signal, consistent with Ziel et al. [29], who documents the predictive value of the earlier-closing EXAA auction for the German day-ahead market. In the current specification, adding the full set of fundamental and reserve features on top of EXAA does not improve the 12:00 point forecast. The EXAA-only model is statistically indistinguishable from the full 12:00 specification, with a GW p-value of 0.65. This matches Eiser and Kleinebrahm [15], who likewise found the compact EXAA-only model to be the best-performing specification for this market.

Table 7: Selected cutoff-grid price forecast results for RQ3.

Forecast specification	RMSE	MAE	n	$p_{\text{GW}}^{07\text{ a}}$	$p_{\text{GW}}^{\text{prev b}}$
07:00	32.63	18.34	17,184	—	—
08:00	32.83	18.31	17,184	1.00	1.00
09:00	32.39	18.25	17,184	0.42	0.43
10:00	31.78	17.81	17,184	0.083	0.091
11:00	30.81	17.53	17,184	0.054	0.14
12:00	27.58	13.72	17,184	< 0.001	< 0.001
12:00, EXAA-only	27.13	13.62	17,184	< 0.001	0.65

^a One-sided GW p-value, squared-error loss, against the 07:00 forecast.

^b One-sided GW p-value, squared-error loss, against the preceding row. For the EXAA-only model, this is the full 12:00 specification.

Errors in EUR/MWh.

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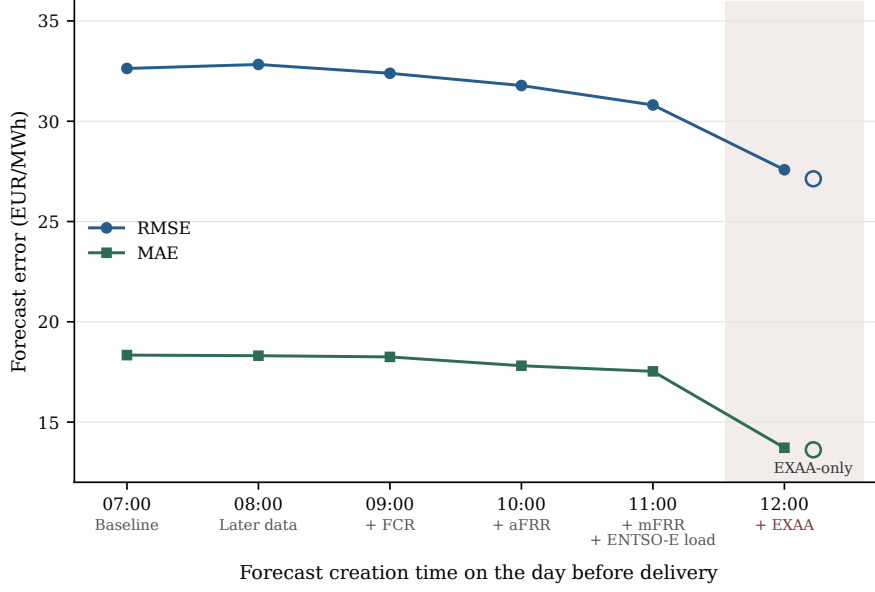


Figure 8: Price forecast accuracy across the forecast creation times of the cutoff grid. The labels below each time name the information added at that cutoff, and realized data advance by one hour at every step. The shaded area marks the 12:00 cutoff, at which EXAA prices become admissible. The hollow markers show the compact EXAA-only benchmark.

H3 is therefore only partly supported. Later and richer information sets lower the price forecast error, and the RMSE at 11:00 is 5.6% below the 07:00 baseline, but before the EXAA price becomes available none of these gains is statistically significant. The only significant improvement, also after the Holm correction, is the step to 12:00. At that cutoff the EXAA price captures most of the information relevant for the point forecast, so the compact EXAA-only model matches the full specification, a known effect for this market rather than a contribution of this thesis.

4.4 Diagnostic Feature Importance

The boosted-tree price forecasts are also analyzed with the SHAP values introduced in Section 3.6.3. The forecast experiment refits a separate LightGBM model for each delivery day on its rolling training window. For this diagnostic, and for each selected specification, these daily models are refitted for every delivery day in the evaluation window with the same training window and hyperparameters, and their SHAP contributions are computed for all MTUs with LightGBM’s built-in TreeSHAP decomposition, then aggregated by the feature groups defined in Section 3.6.3. Because the price model is fitted on the variance-stabilized target of Section 3.4.5, the contributions live in that transformed space and are interpretable in direction and relative magnitude rather than as EUR/MWh amounts. The grouped shares reported here are unit-free ratios of each group’s contribution to the total, so they are comparable across models but describe relative importance in the transformed space, not EUR/MWh contributions. They are diagnostic relative importances within the fitted model, not causal effects.

Figure 9 reports the grouped contribution shares for the RQ2 price models. In the base model,

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raw weather and price lags account for most of the contribution mass, with shares of 43.1% and 35.4%, respectively, and the ENTSO-E load forecast contributes 21.4%. Once OBTF forecasts are added, the structure of the model changes: the OBTF residual-load forecast becomes the most important group, with a contribution share of 42.3% against 24.7% for price lags, while raw weather falls to 13.0% and the OBTF renewable-generation forecast contributes 15.3%. In the weather-free variant, residual load and renewable generation absorb the weather-driven information directly, at 45.4% and 21.7%. This supports the interpretation from RQ2 that the OBTF load and renewable-generation forecasts provide a compact weather-informed representation of the power-system state.

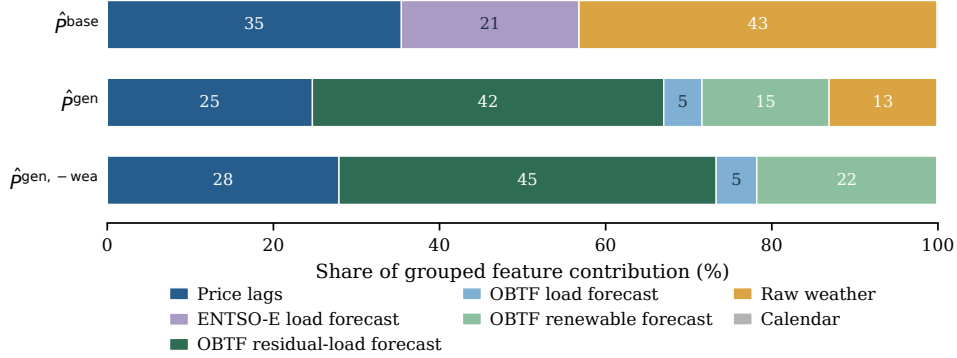


Figure 9: Grouped feature-contribution shares for the RQ2 price models, as percentages of the mean absolute grouped SHAP contribution. The diagnostic covers all delivery days and MTUs in the evaluation window. Shares may not sum exactly to 100 because of rounding.

Figure 10 reports the same diagnostic for selected cutoff-grid models. Before EXAA prices become available, the residual-load forecast and the price lags drive the price forecasts, alongside the renewable-generation forecast, raw weather, and a growing reserve-market block. The reserve-market share rises once aFRR information is available, from 4.4% at 09:00 to 18.5% at 10:00, and to 20.3% at 11:00 as mFRR is added. The fitted models therefore make substantial use of the reserve-market results, even though their contribution to accuracy is not robustly significant (Section 4.3). At 12:00, EXAA prices dominate the feature contributions, at 66.9% of the mass, although all other feature blocks are available. Together with the statistically indistinguishable accuracy of the full 12:00 model and the EXAA-only model, this is consistent with the late EXAA signal already containing most of the short-horizon price information that the other feature blocks provide. Appendix A3.3 reports the leading subgroups for $\hat{P}^{\text{gen}}$ and the 12:00 model, and shows the direction of the leading effects.

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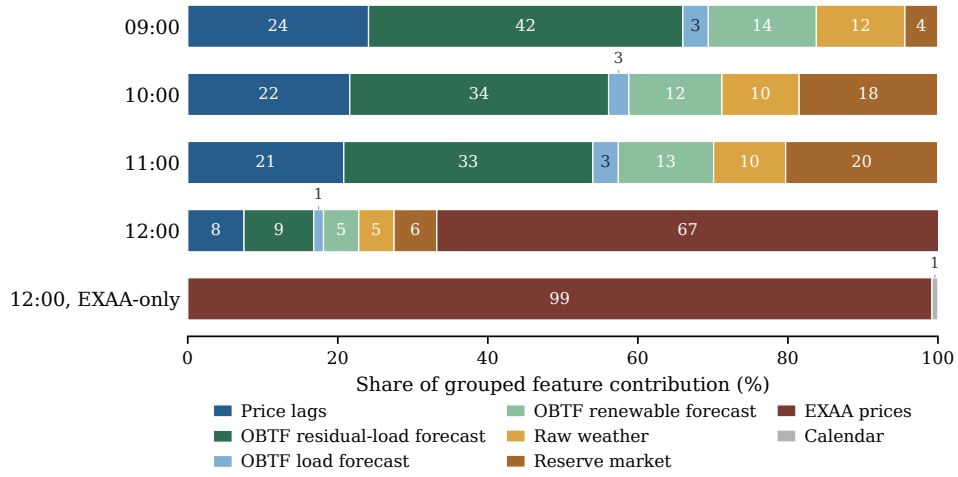


Figure 10: Grouped feature-contribution shares for selected RQ3 cutoff models, as percentages of the mean absolute grouped SHAP contribution. The calendar group is below 1% throughout. The diagnostic covers all delivery days and MTUs in the evaluation window. Shares may not sum exactly to 100 because of rounding.

5 Discussion

5.1 Do OBTF Load and Renewable Forecasts Improve Price Forecasts?

The OBTF forecasts improve the price forecast. Replacing the published ENTSO-E load forecast with the OBTF load, combined renewable-generation, and residual-load forecasts lowers the day-ahead price RMSE by 11.2% relative to the base price forecast, and the improvement is statistically significant. Both forecasts use the same estimator, the same forecast creation time, and the same price-lag, calendar, and raw-weather inputs. They do not, however, share the same information set. The OBTF models draw on a much richer weather representation, installed capacity from the MaStR, population weights, and realized load, which the base model does not receive directly, because hundreds of such features could not be estimated from 70 daily training observations. The gain therefore measures the value of this information delivered in the compact form of explicit forecasts, not the value of the forecast representation alone (Section 5.5). The OBTF load and renewable-generation forecasts thus carry predictive information that the price model does not recover on its own from price history, raw weather, and the public load forecast. This agrees with Maciejowska et al. [8], who show that including load, wind, and solar forecasts, and improving their accuracy, yields more accurate day-ahead price forecasts. Their comparison is against price models without these forecasts, whereas the base forecast here already contains raw weather, so the result extends their finding to a price model that already has access to the underlying weather information.

The weather-free variant refines this interpretation. A price model that keeps the OBTF forecasts but drops the raw weather features is statistically indistinguishable from the model that retains both, and it still improves on the base forecast by essentially the same margin. Once the OBTF forecasts are included, the two-cluster raw weather features thus add no detectable accuracy. This is consistent with how the forecasts are constructed, because each is produced by a task-specific model that already distills the relevant weather signal into an expected load, solar, or wind quantity. For the data-limited price model, these few summary variables appear to be a more usable form of the weather information than the raw weather from which they are derived, although a non-significant difference does not prove that the raw weather is redundant.

The most operationally relevant finding emerges when this result is read together with RQ1. The OBTF solar and wind forecasts are less accurate than the corresponding ENTSO-E benchmarks, yet they still improve the price forecast, whereas the more accurate ENTSO-E renewable-generation forecasts, which several price studies use as direct inputs [11, 12, 14], cannot be used at all because they are published only after gate closure. For an operational price forecast, accuracy and admissibility are therefore both required. A more accurate benchmark that is not observable before the deadline cannot serve as an input, and an admissible forecast is useful only if it adds information beyond price history, weather, and the public load forecast. The OBTF forecasts satisfy both conditions, and the OBTF load forecast additionally exceeds the accuracy of the public forecast it is compared to.

5.2 Operational Value of Waiting

RQ3 shows that the accuracy of the operational price forecast improves as the morning progresses and the admissible information set grows, but gradually and, before 12:00, without statistically significant steps. Between 07:00 and 09:00 the forecast barely changes. From 10:00, when the aFRR results become available, and 11:00, when the public load forecast and the mFRR results enter, the error declines more clearly and is 5.6% lower at 11:00 than at 07:00. On the six-month test period, however, these gains are not statistically significant, which may partly reflect that the backtest uses the same weather run at all cutoffs (Section 5.5). A participant who must commit early therefore loses little by forecasting at 07:00 rather than waiting until 11:00, although the direction of the evidence favors waiting.

A further and larger gain comes at the gate-closure cutoff, once the EXAA price is admissible. The EXAA auction clears earlier than the SDAC auction and its price is a direct market estimate of the same day-ahead quantity, so it is unsurprising that it carries more short-horizon information than any fundamental input, consistent with the predictive value of the EXAA auction documented by Ziel et al. [29]. That a compact model using the EXAA price alone matches the full specification is a known property of this market rather than a finding of this thesis [15], and the feature-importance analysis reflects it, with the EXAA price accounting for about two-thirds of the contribution mass at 12:00.

This does not diminish the value of the OBTF forecasts but clarifies where each source matters. The EXAA price is published only shortly before gate closure, so throughout the earlier part of the morning, while a participant is still forming a bid rather than submitting it at the deadline, no market price is yet available. Across those cutoffs the forecast rests on price lags, weather, the OBTF load and renewable-generation forecasts, and the reserve-market results, which lower the error at every cutoff before 12:00, although not by a robustly significant margin. Once the EXAA price arrives, it dominates.

5.3 Role of Weather and Public Inputs

The literature review left open how weather is best represented in a day-ahead price model. The weather-free variant and the feature-importance results help answer this question. In the base model, raw weather is one of the two largest groups of inputs, so the fitted model relies heavily on weather. When the OBTF forecasts are added, the importance of raw weather drops sharply and is taken over by the OBTF residual-load and renewable-generation forecasts, and the weather-free variant shows that the raw weather features can be removed altogether without a measurable loss. For this data-limited price model, the OBTF forecasts are therefore at least as useful a representation of the weather-driven system state as the two-cluster raw weather fields, plausibly because they condense a much richer weather signal into a small number of variables that are already aligned with how weather affects prices. The dominance of the OBTF residual-load forecast is consistent with residual demand as a central driver of the price [6, 5] and with the explainable machine-learning analysis of Trebbien et al. [20], who find load, wind, and solar generation to be the most important features for German day-ahead prices. This contrasts with operational pipelines that instead feed spatially resolved weather directly into the price model in place of the

unavailable renewable-generation forecasts [15], and with studies that select among a large set of weather variables within the price model [9]. Meteorological forecasts have been shown to improve German price forecasts beyond the day-ahead horizon [10]. At the day-ahead horizon, the results here suggest that explicitly forecasting load and renewable generation is a promising way to bring weather information into a day-ahead price model, subject to the caveat on the unequal weather information discussed in the limitations.

The public data sources play distinct and complementary roles. The published ENTSO-E load forecast is a useful baseline whose accuracy the OBTF load model improves on by correcting it with later information, so in this setting the public forecast is more useful as a starting point for correction than as a final input. The reserve-market results add some information about system tightness before any market price is available, although their effect is not robust on this test period. The EXAA price is the single most informative public input at gate closure, but it is admissible only at the very end of the morning. Read together, these inputs form a sequence in which fundamental and reserve information dominate early and the exchange price dominates late, and an operational price model benefits from using each of them at the point in the morning where it becomes available.

5.4 Openness and Operational Deployment

A defining property of the pipeline is that it is fully open and operationally admissible. Every input is drawn from open, automatically retrievable sources, namely open weather forecasts, the ENTSO-E Transparency Platform, the MaStR, and gridded population data, so the self-generated load, solar, wind, and price forecasts can be generated and extended without proprietary inputs. The preprocessing, the OBTF models, and the price model are released as a modular, configuration-driven codebase that is ready for operational use and deployment, so each stage can be reused and extended. Together with the daily Energy-Arena submission, this follows the call of Lago et al. [2] for open-access benchmarks in electricity price forecasting. By replacing the post-gate-closure ENTSO-E renewable-generation forecasts with OBTF forecasts, the pipeline also respects the operational timing, so the reported accuracy reflects what a participant could actually have produced before the deadline.

The pipeline is additionally submitted to Energy-Arena [18], where its load, solar, wind,² and price forecasts are scored against realized outcomes under a common gate-closure deadline and shared in the benchmark’s cooperative mode so that other forecasters can reuse them, for example as ensemble members. This provides a continuous, forward-looking, and leakage-controlled evaluation that extends the fixed test period of this thesis and lets the OBTF forecasts serve as a reusable building block for further work.

Beyond serving the price model, the OBTF component forecasts are directly reusable. Regenerated from the open repository, the OBTF load forecast improves on the public ENTSO-E load forecast and can be used in its place, while the OBTF solar and wind forecasts provide admissible substitutes for the ENTSO-E renewable-generation forecasts, which are not published before gate

²The Energy-Arena DE-LU wind challenge targets onshore wind generation, so only the onshore component of the OBTF wind forecast is submitted there, whereas the OBTF wind forecast evaluated in this thesis is total wind generation, onshore and offshore.

closure. Unlike the ENTSO-E forecasts, which are produced by undisclosed models and cannot be inspected, the OBTF forecasts are fully transparent, because their inputs, models, and construction are open and can be examined and adapted rather than taken as given.

5.5 Limitations

The load benchmarking in RQ1 is an operational rather than a like-for-like modeling comparison. The OBTF load forecast corrects the public ENTSO-E forecast with information available up to the pre-gate-closure cutoff, whereas the ENTSO-E forecast is produced by an undisclosed model at a generally earlier, unspecified cutoff. A comparison that fixes both forecasts to an identical cutoff is therefore not feasible, and the reported load improvement should be read as the operational value of correcting the public forecast before gate closure rather than as evidence of a superior standalone load model. This caveat concerns only the interpretation of the standalone load comparison. Because the OBTF load forecast serves solely as an admissible input to the price model, its only operational requirement is to be available before the 12:00 gate closure, so producing it at a later cutoff than the ENTSO-E forecast, with correspondingly more recent information, is by design rather than a disadvantage.

The central caveat of the RQ2 comparison, discussed in Section 5.1, is that the base price model and the OBTF models do not share the same information set. The base model receives raw weather only through a compact two-cluster aggregation of the ICON-D2 fields and none of the capacity, population, and realized-load inputs of the OBTF models. A comparison that supplied the base model with all of these inputs directly would isolate the effect of the explicit representation, but hundreds of such features cannot be estimated reliably from 70 daily training observations. The OBTF forecasts avoid this by splitting the problem. Load, solar, and wind generation depend on different weather variables, such as temperature for load, irradiance and cloud cover for solar, and wind speed for wind. Without the first-stage forecasts, a single price model would have to receive all of these variables at once. With them, each OBTF model uses only the weather it needs, and the price model receives the result as a small number of forecast series, namely load, renewable generation, and residual load. This limitation is thus at the same time the central appeal of the approach.

The evaluation covers a single test window from February to July 2026, which is the period for which post-transition quarter-hourly prices and the full set of operational inputs are available. The results therefore describe one half-year of a market that only recently moved to quarter-hourly clearing, and the autumn and mid-winter months lie outside the test window. Whether the reported effects are stable across seasons and across years with different fuel-price and renewable-capacity conditions cannot be established from this sample, and a longer multi-year evaluation would be needed to settle it. The ongoing Energy-Arena submission extends this evidence with each new delivery day.

A major limitation of RQ3 concerns the weather input. The earlier ICON-D2 runs that would be admissible at the 07:00 to 09:00 cutoffs are not archived for the whole test period, so all cutoffs use the 06 UTC run, which is available only from about 09:20. The early cutoffs therefore receive a more recent weather forecast than they could use in operation, which favors them. At the same

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time, the backtest cannot capture any improvement from newer weather runs over the morning, although weather is a central input of both the OBTF and the price models. Both effects bias the measured gain from waiting downward and may explain why the gains before 12:00 are not statistically significant. A backtest with the operational run schedule of Table 4, possible once the earlier runs have been archived over a full test period, would resolve this.

The OBTF solar and wind forecasts remain less accurate than the ENTSO-E benchmarks, and the feature-importance analysis attributes more of the price model’s contribution to the OBTF load and residual-load forecasts than to the OBTF renewable-generation forecast (Section 4.4). The OBTF solar and wind forecasts are thus admissible and useful but not competitive in absolute accuracy, and closing this gap is a limitation of the current OBTF models rather than a property of the price model. The capacity data from the current MaStR also include late-registered units that were not yet known at the time, an effect the rolling bias correction partly absorbs. In addition, the archived ICON-D2 irradiance is incomplete or implausible on a few days in June 2026. These days are retained, so the affected solar forecasts, and through them the price forecasts, carry these input errors. As with the OBTF load forecast, however, the comparison is not on equal footing. The ENTSO-E solar and wind benchmarks are produced by undisclosed models and, being published only after gate closure, may draw on later weather runs and more recent information than the OBTF forecasts, which are restricted to the pre-gate-closure cutoff. Part of the observed gap may therefore reflect this information and timing advantage rather than modeling skill alone, and how much of it is genuinely attributable to the OBTF models cannot be established from this comparison.

The feature-importance analysis is diagnostic rather than causal. The SHAP contributions describe how the fitted model distributes its predictions across inputs on the sampled data, not the true causal effect of each input, and they are computed on the variance-stabilized model target so that only their direction and relative magnitude are interpretable.

Finally, the thesis evaluates point forecasts only. Operational trading and risk management typically require the full predictive distribution, and the pipeline does not yet produce or evaluate probabilistic price forecasts.

5.6 Future Work

Several extensions follow directly from these limitations. The most immediate is to close the accuracy gap of the OBTF solar and wind forecasts, for example through multi-model weather ensembles [48], wind speeds interpolated to the turbine-specific hub heights recorded in the MaStR, or an explicit treatment of curtailment, since more accurate renewable-generation forecasts would strengthen the fundamental contribution to the price model.

The estimators themselves could also be revisited. Tabular foundation models such as TabPFN are designed for small training sets [49], which makes them a natural candidate for the data-limited price model with its 70-day training window.

Beyond point forecasts, the pipeline could be extended to probabilistic forecasting. The OBTF models could provide quantile forecasts of load, solar, and wind generation as price-model inputs, which Uniejewski and Ziel [50] show to lower the price RMSE by up to 13% compared with

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point inputs for the German market. The same OBTF forecasts and cutoff structure could feed a distributional or quantile price model [12], which would make the operational value of the approach measurable in terms of the full predictive distribution rather than the conditional mean alone.

Further gains may come from a broader admissible information set. Cross-border prices and residual-load forecasts of neighboring bidding zones, as well as fuel and emission-allowance prices, are known to carry predictive information for German day-ahead prices [11, 14, 5], and incorporating them within the open-data and pre-gate-closure constraints of this thesis would show how much further the operational forecast can be improved.

Finally, a longer and multi-seasonal evaluation would establish the stability of the reported effects across market regimes, and extending the setup to the intraday markets that follow the day-ahead auction would broaden its operational relevance.

6 Conclusion

The main contribution of this thesis is a set of fully operational forecasting models for load, solar, and wind generation based exclusively on open data, the OBTF models. They are released as part of an open pipeline that enforces forecast-time information availability and that others can rerun and reuse. Their daily submission to the Energy-Arena benchmark [18] lets anyone verify their accuracy beyond the test period of this thesis. In contrast to the ENTSO-E forecasts, whose methods are not disclosed, their inputs and construction are fully open. Each model is either more accurate than its ENTSO-E counterpart, as for load, or admissible before gate closure when the ENTSO-E forecast is not, as for solar and wind. Used as explicit inputs, the OBTF forecasts improve the operational day-ahead price forecast for the DE-LU bidding zone, in which every input must be available before the 12:00 gate closure. The models can therefore be reused as admissible, transparent inputs in other operational applications.

For RQ1, the OBTF load forecast is more accurate than the published ENTSO-E forecast in the operational setting, reducing the load RMSE by 36.0%. The OBTF solar and wind forecasts are less accurate than the ENTSO-E forecasts, but those benchmarks are not published before gate closure and cannot be used in an operational price forecast.

For RQ2, adding the OBTF load, combined renewable-generation, and residual-load forecasts to the price model lowers the day-ahead price RMSE by 11.2%, a statistically significant improvement. The weather-free variant is statistically indistinguishable in accuracy, so the raw weather fields add no detectable accuracy once these forecasts are included, which makes them a compact representation of the weather-driven system state for the price model.

For RQ3, price accuracy improves as the morning progresses, but gradually. Up to 11:00 the error falls by 5.6% without a statistically significant step, which may partly reflect that the backtest uses the same weather run at all cutoffs. The only significant gain follows at gate closure once the EXAA price becomes admissible. That the earlier-clearing EXAA price captures most of the achievable accuracy at 12:00 is a known property of this market rather than a result of this thesis [15]. Before any market price is available, the forecast rests on price lags, weather, the OBTF forecasts, and the reserve-market results.

The findings are relevant beyond the market and period studied here. Day-ahead prices depend strongly on weather-driven load and renewable generation [7, 5], and the corresponding public forecasts are either less accurate than the OBTF forecast or unavailable before gate closure. Turning open weather data into admissible load and renewable-generation forecasts is therefore valuable in other bidding zones with the same gate-closure constraint. In a field where forecasts are often produced by undisclosed models and evaluated retrospectively, such an open and continuously benchmarked pipeline offers a transparent operational baseline that others can reproduce, extend, and compare against. The thesis is thus a step toward day-ahead price forecasting that is operational, open, and explicit about what information it uses and when.

Appendix A

A1 Data Availability and Operational Timing

Table A1 documents the main availability times shown in Figure 3. Regulatory deadlines and exchange rules are binding latest times. Platform behavior and publication latencies describe typical availability and can deviate on individual days. The admissible information set of Section 3.3 is applied with these availability times at every cutoff.

Table A1: Sources and status of the operational availability times used in Figure 3.

Element	Operational timing used	Source and status
SDAC gate closure	12:00 on day $d - 1$	Exchange rule [3]
ENTSO-E day-ahead load forecast	Regulatory deadline two hours before gate closure, observed available around 10:30 on day $d - 1$ in the data collection	Regulatory deadline, Regulation 543/2013 Art. 6(1)(b) [16], availability observed by the data collection
ENTSO-E realized load	Published no later than one hour after the operating period	Regulatory deadline, Art. 6(1)(a) [16]
ENTSO-E realized wind and solar generation	Published no later than one hour after the operating period, with initial estimates updated as measured data becomes available	Regulatory deadline, Art. 16(1)(j) [16]
ENTSO-E day-ahead wind and solar forecast	Published after the 12:00 gate closure and no later than 18:00 on day $d - 1$	Publication time reported for the ENTSO-E Transparency Platform [18], with the publication requirement set under Art. 14(2)(d) [16]
EXAA day-ahead prices	Auction closes at 10:15 on day $d - 1$. EXAA day-ahead prices for Germany are published on the ENTSO-E Transparency Platform around 11:15	Exchange rule [31], publication time [17, 15]
Control-reserve auction results	German control-reserve capacity auctions for delivery day d close on day $d - 1$ at 08:00 (FCR), 09:00 (aFRR), and 10:00 (mFRR). Their results are used from one hour after gate closure. A check on 28 September 2026 found them published 14 to 23 minutes after gate closure	Auction platform [32]
SDAC auction results	Released after gate closure and therefore not admissible before 12:00	Market-coupling process [4]
ICON-D2 weather forecasts	A new run starts every three hours and is complete about 80 minutes after its start	DWD Open Data [25] and Open-Meteo documentation [26]

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A2 Model Configurations

Table A2 summarizes the final configuration of every model in the pipeline. All models use a rolling origin: for each delivery day they are refitted on a trailing training window ending on the day before delivery. The load, solar, and wind forecasting models use their own task-specific weather representations and forecast creation times, while the price models use the coarse two-cluster weather aggregation selected on the development period (Appendix A3.1). The RQ3 cutoff models reuse the RQ2 price-model configuration and vary only the admissible information set with the forecast creation time.

Table A2: Final configuration of each pipeline model. The window is the length of the rolling training window in days. All weather features are derived from the DWD ICON-D2 model, accessed through DWD Open Data or the Open-Meteo API.

Model	Estimator	Win.	Weather representation	Key inputs
Load, correction	LightGBM	224	ICON-D2, 25 clusters, population-weighted	ENTSO-E load forecast, realized load to 10:30, calendar, weather
Load, direct	LightGBM	224	ICON-D2, 25 clusters, population-weighted	Realized load to the cutoff, calendar, and weather, without the ENTSO-E forecast
Solar	Hist. gradient boosting	90	ICON-D2, 25 clusters per TSO, capacity-weighted	Clear-sky geometry and physics baseline, irradiance, cloud cover, installed capacity
Wind	Hist. gradient boosting	180	ICON-D2 incl. hub-height fields, 100 clusters incl. offshore, capacity-weighted	Hub-height wind speed, power-curve and cubic transforms, installed capacity
Price $\hat{P}^{\text{base}}$	LightGBM, 300 trees	70	ICON-D2, 2 clusters	Price lags, calendar, weather, ENTSO-E load forecast
Price $\hat{P}^{\text{gen}}$	LightGBM, 300 trees	70	ICON-D2, 2 clusters	As $\hat{P}^{\text{base}}$, with the ENTSO-E load forecast replaced by the OBTF load, renewable-generation, and residual-load forecasts
Price, RQ3 cutoffs	LightGBM, 300 trees	70	ICON-D2, 2 clusters, 06 UTC run at all cutoffs	Inputs admissible at each cutoff, adding reserve-market results from 09:00 and EXAA prices at 12:00

A3 Additional Price Forecast Results

This appendix reports the development-period selection of the base price model, which fixes the rolling training window and the weather-cluster count used throughout the price experiments.

A3.1 Selection of the Base Price Model Window and Weather Clustering

The rolling training window and the raw-weather cluster count of the base price model are selected by a staged search on the development period (20 December 2025 to 31 January 2026), so that no design choice depends on the test period. The search is a coordinate descent. The training window is chosen first at a fixed mid-range count of sixteen weather clusters, the boosted-tree hyperparameters are chosen next at the selected window, and the number of weather clusters is chosen last at the selected window. The hyperparameter stage selects a small tree size and moderate L_2 regularization, with the number of leaves having little effect at the available sample size and stronger regularization degrading accuracy. The two knobs that carry the comparison are therefore the window length and the cluster count, reported below.

Table A3 varies the training window from 28 days to an expanding window that uses all post-transition history. Development-period accuracy improves up to 70 days and then plateaus, and the expanding window gives no meaningful further gain. A 70-day window is selected.

Table A3: Development-period accuracy of the base price model over the rolling training window, at sixteen weather clusters and 300 boosted-tree rounds. The expanding window uses all post-transition history. The selected window is in bold. Errors are in EUR/MWh.

Training window (days)	RMSE	MAE
28	26.75	16.79
42	26.37	16.36
56	24.49	14.75
70	23.30	14.19
Expanding	23.21	14.28

Table A4 fixes the window at 70 days and varies the number of weather clusters from 1 to 100. Accuracy is best at two clusters and degrades as the spatial resolution increases, while the mean training time per delivery day grows from about twelve seconds at one cluster to nearly half an hour at one hundred clusters. Finer spatial resolution therefore raises the training cost sharply without improving accuracy, so no accuracy–runtime trade-off arises from using a coarse grid. The selected two-cluster representation is the most accurate and among the cheapest, with only the single-cluster grid marginally faster but less accurate. A plausible explanation is that the data-limited price model, which sees at most 70 daily training rows, cannot exploit fine spatial weather detail and instead overfits it. Two clusters are selected. To keep the sweep tractable given this steep cost growth, the cluster search uses 100 boosted-tree rounds rather than the 300 rounds of the final model, which is why the same sixteen-cluster, 70-day configuration appears at a lower error in Table A3, run at 300 rounds, than in Table A4. The selection assumes that using fewer rounds shifts the error level roughly uniformly rather than reordering the cluster counts. This assumption is not tested for every count, but the selected two-cluster configuration is retrained at the full 300 rounds for all reported results. Figure A1 shows the same result as an accuracy–runtime frontier. The selection is robust to the choice of metric. The two-cluster representation minimizes both the RMSE and the MAE, and the 70-day window has the lowest MAE of all windows while lying on

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the RMSE plateau.

Table A4: Development-period accuracy and mean training cost of the base price model over the number of weather clusters, at a 70-day window and 100 boosted-tree rounds. Training time is the mean wall-clock time per delivery day of the rolling backtest. The selected cluster count is in bold. Errors are in EUR/MWh.

Weather clusters	RMSE	MAE	Training time (s/day)
1	24.21	14.89	12
2	23.70	14.49	17
5	24.02	14.86	31
8	23.96	14.63	49
12	24.31	14.92	79
16	24.33	14.95	110
25	24.56	15.03	189
40	25.43	15.61	363
64	25.12	15.58	758
100	25.49	15.64	1712

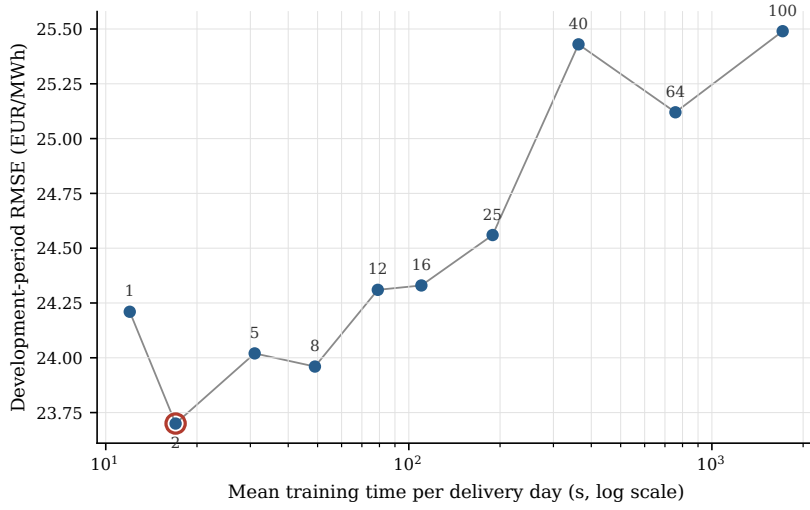


Figure A1: Development-period RMSE against mean training time per delivery day for the base price model, over weather-cluster counts from 1 to 100 at a 70-day window. The selected two-cluster representation is highlighted.

A3.2 Reserve-Market Ablation

Table A5 isolates the effect of the reserve-market features by comparing each cutoff price model with an otherwise identical model that excludes the reserve-market covariates, on the same matched panel. Before the EXAA price is admissible, adding the reserve-market results lowers the RMSE at every cutoff, by 1.1% at 09:00, 3.4% at 10:00, and 2.1% at 11:00. Only the 10:00 improvement, after the aFRR results enter, is significant on its own, and it does not remain significant after the

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Holm correction across the three pre-EXAA cutoffs. At the 12:00 cutoff, once the EXAA price is admissible, the reserve-market features no longer help and slightly increase the error, which is consistent with the EXAA signal absorbing most of the short-horizon price information. The reserve-market results therefore contribute to the gradual improvement before 12:00, but their effect cannot be established robustly on this test period.

Table A5: Reserve-market ablation for the cutoff price models. Each row compares a cutoff model without reserve-market features against the otherwise identical model with them, on the matched test panel from 1 February 2026 to 31 July 2026 ($n = 17,184$). The *With reserve* RMSE column holds the selected cutoff models and matches their RMSE in Table 7, while the *No reserve* column is the counterfactual without reserve-market features. Errors are in EUR/MWh.

Cutoff	Reserve products	RMSE		Δ RMSE	p_{GW}^a
		No reserve	With reserve		
09:00	FCR	32.76	32.39	-1.1%	0.23
10:00	FCR, aFRR	32.89	31.78	-3.4%	0.025
11:00	FCR, aFRR, mFRR	31.46	30.81	-2.1%	0.11
12:00 ^b	FCR, aFRR, mFRR	27.27	27.58	+1.1%	—

^a One-sided GW p-value, squared-error loss, with against without reserve-market features. None remains significant after a Holm correction across the three pre-EXAA cutoffs.

^b Both 12:00 models include the EXAA price. The reserve-market features do not improve the forecast, so no p-value is reported.

A3.3 Detailed Feature-Importance Results

The grouped shares in Section 4.4 aggregate over feature groups. A subgroup collects all the per-MTU columns of one signal, such as the lagged price on day $d - 1$, which enters as a separate feature for each of the 96 quarter-hours, or the OBTF residual-load forecast. Tables A6 and A7 break the feature groups down one level finer, into these subgroups, for the price forecast with OBTF load and renewable-generation forecasts $\hat{P}^{\text{gen}}$ and for the full 12:00 cutoff model.

Shares use the same normalization as the grouped shares in Figures 9 and 10, so a group with a single subgroup, such as the EXAA price, matches those figures exactly. Subgroups within a group need not sum to the group share, because importance is measured as a mean absolute contribution: when two subgroups in the same group push the forecast in opposite directions on a given prediction, their effects partly cancel in the group total, so the group’s absolute contribution is smaller than the sum of the subgroups’ absolute contributions.

For $\hat{P}^{\text{gen}}$, the residual-load level is the single largest subgroup, followed by the previous-day price lag, with the residual-load ramp, wind, radiation, and the OBTF renewable-generation forecasts contributing further. For the 12:00 model, the EXAA price subgroup alone accounts for about two-thirds of the contribution mass, far ahead of the residual-load level, the previous-day price lag, and the reserve-market results, which confirms the dominance of the late EXAA signal at the final cutoff. Figure A2 shows the direction of the leading effects in $\hat{P}^{\text{gen}}$. Higher recent prices

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and higher residual load push the forecast up, while higher wind generation pushes it down, in the economically expected direction.

Table A6: Subgroups with the largest mean absolute SHAP contribution for $\hat{P}^{\text{gen}}$, combining each signal over its MTUs. Shares are percentages of the model’s total grouped contribution.

Subgroup	Group	Share (%)
Residual load	Residual load	36.9
Lagged price (day $d - 1$)	Price lags	21.2
Residual load ramp	Residual load	9.9
Wind generation	Renewables	9.9
Shortwave diffuse radiation	Raw weather	6.2
Wind speed	Raw weather	5.8
Shortwave direct radiation	Raw weather	5.0
Load forecast	Load	4.6
Solar generation	Renewables	4.5
Total renewable generation	Renewables	4.1

Table A7: Subgroups with the largest mean absolute SHAP contribution for the full 12:00 cutoff model, combining each signal over its MTUs. Shares are percentages of the model’s total grouped contribution.

Subgroup	Group	Share (%)
EXAA price	EXAA	66.9
Residual load	Residual load	6.9
Lagged price (day $d - 1$)	Price lags	6.1
Reserve market	Reserve	5.7
Residual load ramp	Residual load	3.9
Wind generation	Renewables	2.8
Wind speed	Raw weather	2.7
Shortwave diffuse radiation	Raw weather	1.9
Total renewable generation	Renewables	1.7
Shortwave direct radiation	Raw weather	1.6

APPENDIX A

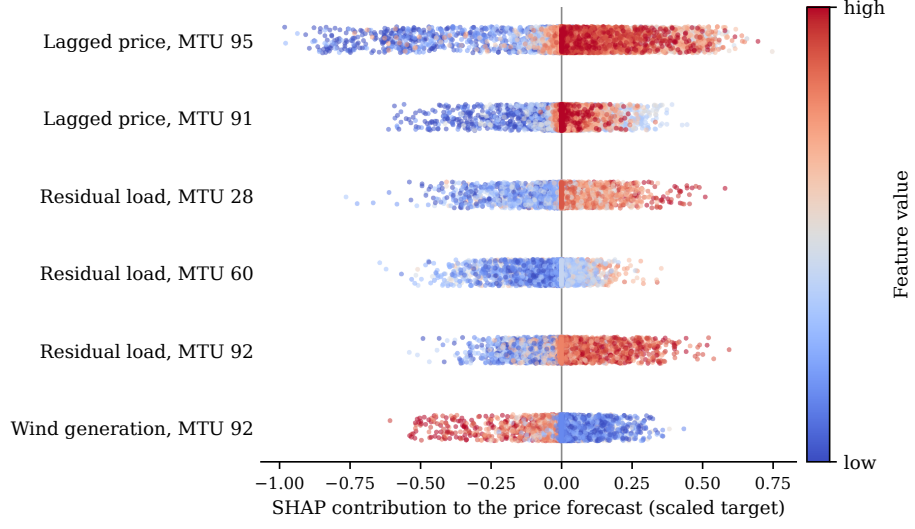


Figure A2: Direction of the leading feature effects for $\hat{P}^{\text{gen}}$. Each point is one sampled prediction, its horizontal position is the feature's SHAP contribution to that forecast, and its color is the feature value from low (blue) to high (red). Contributions are on the scaled model target, so only their direction and relative magnitude are interpretable.

Data and Code

The operational forecasting pipeline developed in this thesis is publicly available at

https://github.com/davideig/DA_Price_Forecasting_DE_LU_Fundamental

It provides the data-access integrations and endpoints, the preprocessing and modeling code, and the configuration files, so that open-data load, solar, wind, and day-ahead price forecasts for the DE-LU market can be generated and extended without building the data infrastructure from scratch.

The resulting load, solar, wind, and day-ahead price forecasts are also submitted to Energy-Arena [18] under the username **Eiglsperger_2026**, using the benchmark’s “open immediately” cooperative mode, so that other forecasters can download them before the target period is realized and reuse them, for example as members of forecast ensembles.

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Karlsruhe, den 30.09.2026

David Maximilian Eiglsperger

Anlage: Erklärung zur Verwendung generativer KI-Systeme

Bei der Erstellung der Arbeit habe ich die folgenden auf künstlicher Intelligenz (KI) basierten Systeme benutzt:

- 1. Claude (Anthropic)
- 2. Claude Code (Anthropic)
- 3. Codex (OpenAI)

Ich erkläre weiterhin, dass ich

- ☒ mich aktiv über die Leistungsfähigkeit und Beschränkungen der oben genannten KI-Systeme informiert habe,
- ☒ überprüft habe, dass die mithilfe der oben genannten KI-Systeme generierten und von mir übernommenen Inhalte faktisch richtig sind,
- ☒ mir bewusst bin, dass ich als Autor dieser Arbeit die Verantwortung für die in ihr gemachten Angaben und Aussagen trage.

Die oben genannten KI-Systeme habe ich wie im Folgenden dargestellt eingesetzt.

Arbeitsschritt	Eingesetzte(s) KI-System(e)	Beschreibung der Verwendungsweise
Generierung von Ideen und Konzeption der Arbeit	Claude	Diskussion möglicher Forschungsfragen und experimenteller Designs als Ausgangspunkt für eigene Entscheidungen
Literatursuche	–	–
Literaturanalyse	Claude	Unterstützung beim Verständnis spezifischer Methoden und Modelle
Literaturverwaltung und Zitationsmanagement	Claude	Überprüfung von Referenzeinträgen auf formale Vollständigkeit. Alle Zitate wurden selbstständig auf Korrektheit geprüft
Auswahl von Methoden und Modellen	Claude, Claude Code	Diskussion geeigneter Prognosemethoden und Modellspezifikationen. Die finale Methodenauswahl wurde selbst getroffen
Datensammlung und -analyse	Claude, Claude Code, Codex	Unterstützung beim Verständnis der ICON-D2-, ENTSO-E- und weiteren Open-Data-Strukturen. Überprüfung der Datenverarbeitungspipeline in Python

Arbeitsschritt	Eingesetzte(s) KI-System(e)	Beschreibung der Verwendungsweise
Generierung von Programmcodes	Claude Code, Codex	Implementierung und Überarbeitung der Skripte und Konfigurationen im Repository. Korrektur von LaTeX-Code. Fehlersuche und Optimierung bestehender Code-Abschnitte
Erstellung von Visualisierungen	Claude, Claude Code	Unterstützung bei Implementierung und Verbesserung von Abbildungen in Python und LaTeX
Interpretation und Validierung	Claude	Diskussion von Modellergebnissen als Denkansatz. Die finale Interpretation und Validierung erfolgten selbstständig
Strukturierung des Texts der Arbeit	Claude	Vorschläge für Gliederung und Kapitelaufbau
Formulierung des Texts der Arbeit	Claude	Erste Formulierungsideen und Umformulierung von Wörtern und Absätzen
Übersetzung des Texts der Arbeit	–	–
Redigieren des Texts	Claude	Rechtschreib-, Grammatik- und Stilprüfung sowie Konsistenzprüfung. Alle Korrekturen wurden selbstständig geprüft und eigenverantwortlich übernommen
Vorbereitung der Präsentation des Texts	–	–
Sonstiges	–	–

Karlsruhe, den 30.09.2026

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